



Analog IC Design

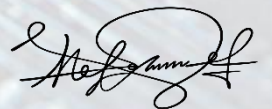
CMOS Technology

MOHAMMED NASR EL-DIN

Full custom op-amp



cadence®

A handwritten signature in black ink, likely of the author, Mohammed Nasr El-Din.

Summary

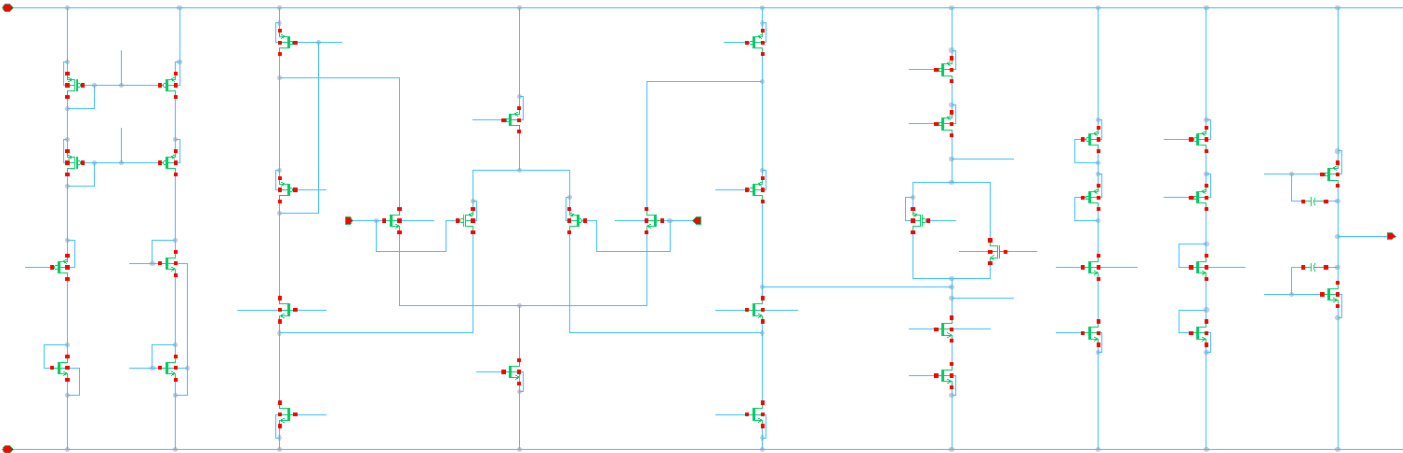
This report presents the design, sizing, and simulation of a full-custom two-stage CMOS operational amplifier, developed as part of the Analog IC Design (CMOS Technology) program at the Information Technology Institute (ITI), in 65nm CMOS technology at a 2.5V supply.

The design targets a DC open-loop gain greater than 60dB, a phase margin greater than 60°, and a gain-bandwidth product greater than 10MHz, verified across all standard process corners and a wide temperature range.

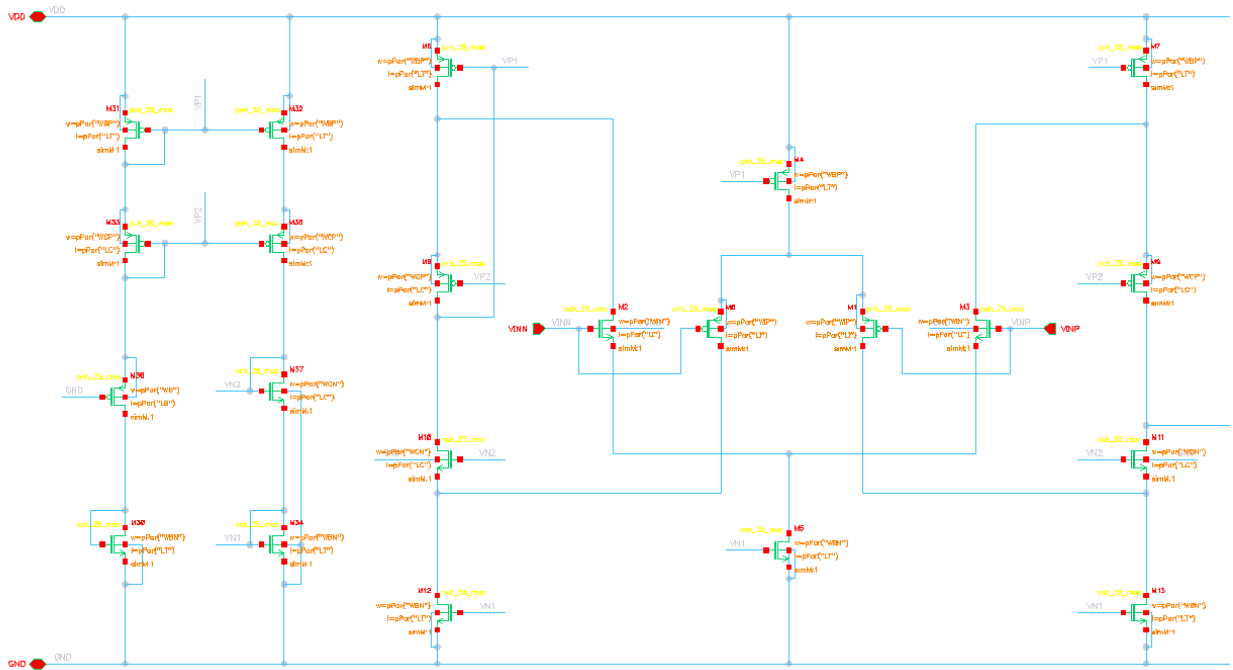
The circuit topology is based on the compact two-stage rail-to-rail CMOS operational amplifier presented in the referred paper: [Rail-to-Rail Input/Output Operational Amplifier for VLSI Cell Libraries by Hogervorst, Tero, Eschauzier, and Huisingsh \(IEEE JSSC, Dec. 1994\)](#), adopting its floating class-AB output stage biased through a Monticelli-type translinear loop, which reduces the noise and offset contribution of the class-AB biasing without needing extra dedicated bias sources.

Transistor sizing was carried out using the g_m/I_D design methodology with the ADT (Analog Designer's Toolbox) tool, following standard trade-offs for the input pair, current sources, and cascode devices. Two bias-generation schemes for the cascode transistors were explored — a shared cascode current mirror, and independent bias branches per node — and both were verified through DC, AC (loop-gain and stability), noise, and transient simulations across process corners.

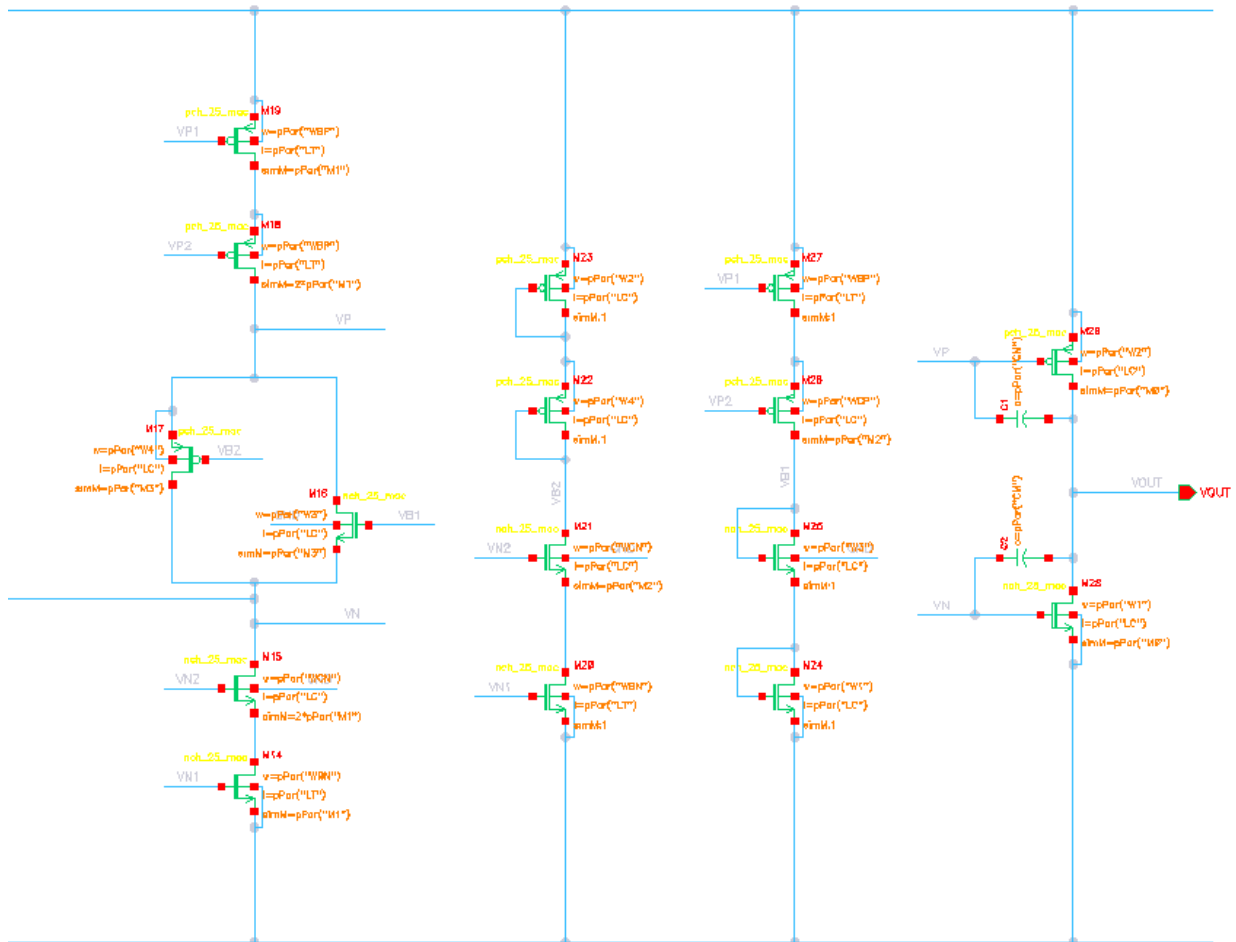
The amplifier was also verified in closed-loop configurations (inverting, non-inverting, and integrator), where the measured response matched the expected theoretical behavior in each case.



Technology	65nm
Supply voltage	2.5 V
corners	Temp: -40°c → 125°c Process: TT, FF, SS, FS, SF
R_L	20 KΩ
C_L	20 pF
A_{ol}	> 60 dB
PM	> 60°
GBW	> 10 MHz



Current source, current mirror, and input stage



Monticelli translinear loop, and class AB output stage

OP-AMP design

We will directly assume values for L and g_m/I_D (or V^*) based on your designer's intuition and folded cascode trade-offs matrix. These assumptions will greatly simplify the design process and can be later tuned in simulations

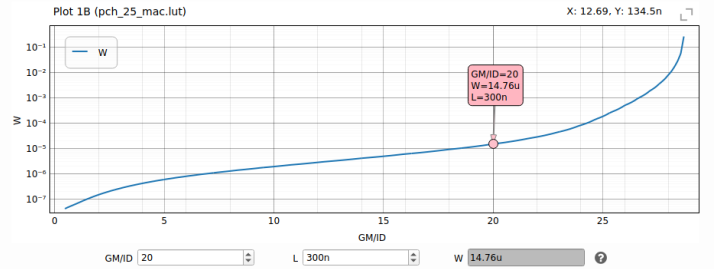
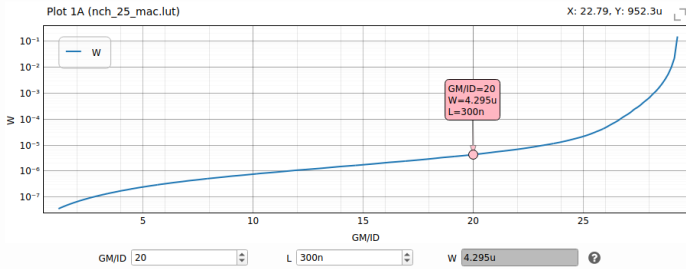
For the input pairs $M_{0,1,2,3}$, assume small $L = 300nm$ and bias it in MI or WI $g_m/I_D = 20\mu S/\mu A$. This maximizes the GBW and minimizes the input capacitive loading.

For current sources, use relatively long $L = 1\mu m$ and bias them in SI $g_m/I_D = 10\mu S/\mu A$, a large g_m will not help the gain but will increase the offset and noise.

For cascode transistors, use moderate $L = 0.5\mu m$ and bias them in MI or WI $g_m/I_D = 15$, a large g_m helps the gain and doesn't increase the noise.

We use a current split of $S = I_{B1}/I_{B2} = 2$ between the input (CS) and folding (CG) branches of the single-ended folded cascode stage. This improves DC gain, bandwidth, and noise density compared to equal splitting ($S=1$), while keeping acceptable slew rate and phase margin. For further details on the effect of the current split ratio and its optimization, refer to the paper. [H. Omran, "Optimum Split Ratio for Folded Cascode OTA Bias Current: A Qualitative and Quantitative Study," Des 2019.](#) (it's preferred to open the link using phone device if it didn't run on computer device)

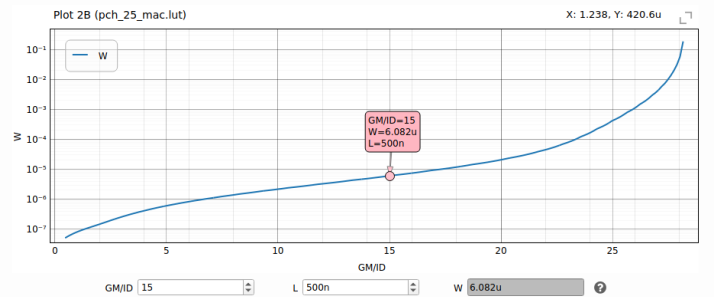
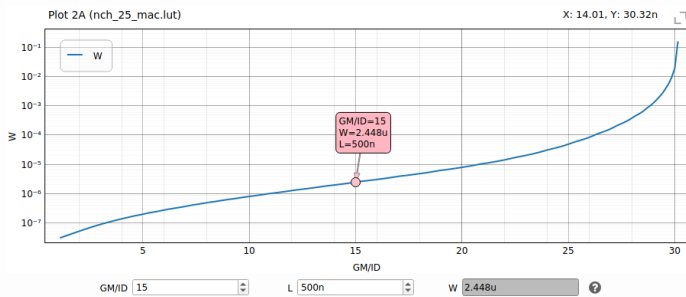
For the input pairs, $g_m/I_D = 20\mu S/\mu A$, $L = 300nm$, let $I_D = 10\mu A$, $V_{DS} = 1.1V$, $V_{SB-NMOS} = 0.75V$



$$W_{IN} = 4.5\mu m, W_{IP} = 15\mu m$$

For the cascode transistors $M_{8,9,10,11}$, they will carry $5\mu A$ current based on the selected current splitting ratio.

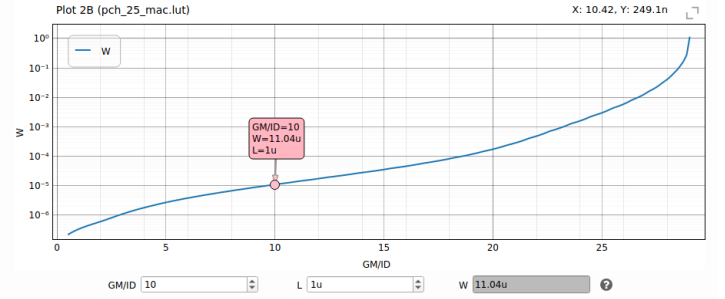
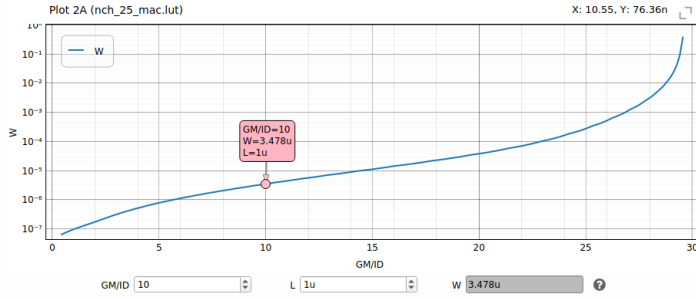
$g_m/I_D = 15\mu S/\mu A$, $L = 0.5\mu m$, let $V_{DS} = 0.7V$, $V_{SB-NMOS} = 0.7V$



$$W_{NMOS} = 2.5\mu m, W_{PMOS} = 6\mu m \text{ for } I_D = 5\mu A$$

For any other cascode device, we can determine its suitable width depending on his drain current by the equations: $W_N = 2.5\mu m \times I_D/5\mu A$ for NMOS devices and $W_P = 6\mu m \times I_D/5\mu A$. The determined width will be implemented by multipliers from unit transistors, that gives better mismatch.

For the current mirror transistors that will carry $10\mu A$ drain current, $L = 1\mu m$ and $g_m/I_D = 10\mu S/\mu A$, $V_{SB} = 0$, let $V_{DS} = 0.9V$

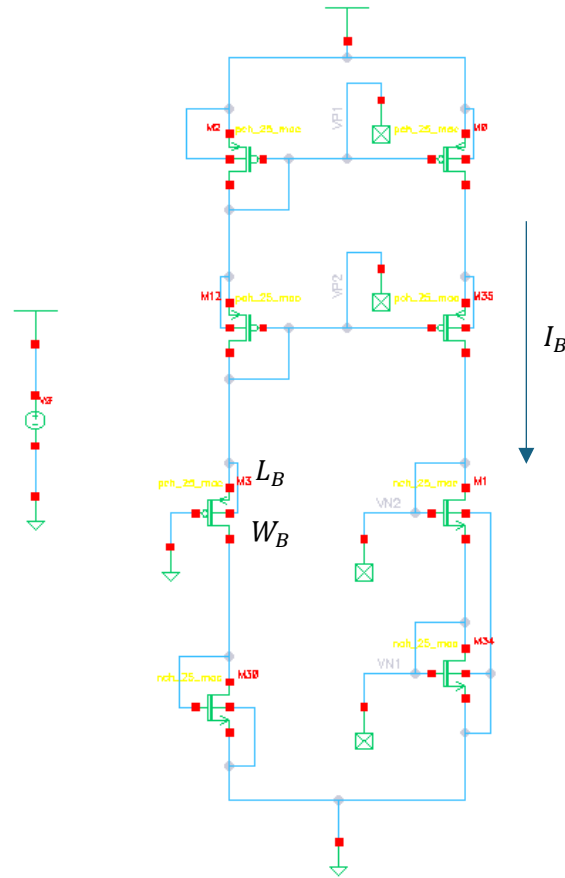


$$W_N = 3.5\mu m, W_P = 11\mu m \text{ for } I_D = 10\mu A$$

Note, the values of the current source's width has been tuned during the simulation to be $W_N = 4.2\mu m$, $W_P = 13.5\mu m$.

For the others current mirrors, we can determine its suitable width depending on his drain current by the equations: $W_N = 4.2\mu m \times I_D/10\mu A$ for NMOS devices and $W_P = 13.5\mu m \times I_D/10\mu A$.

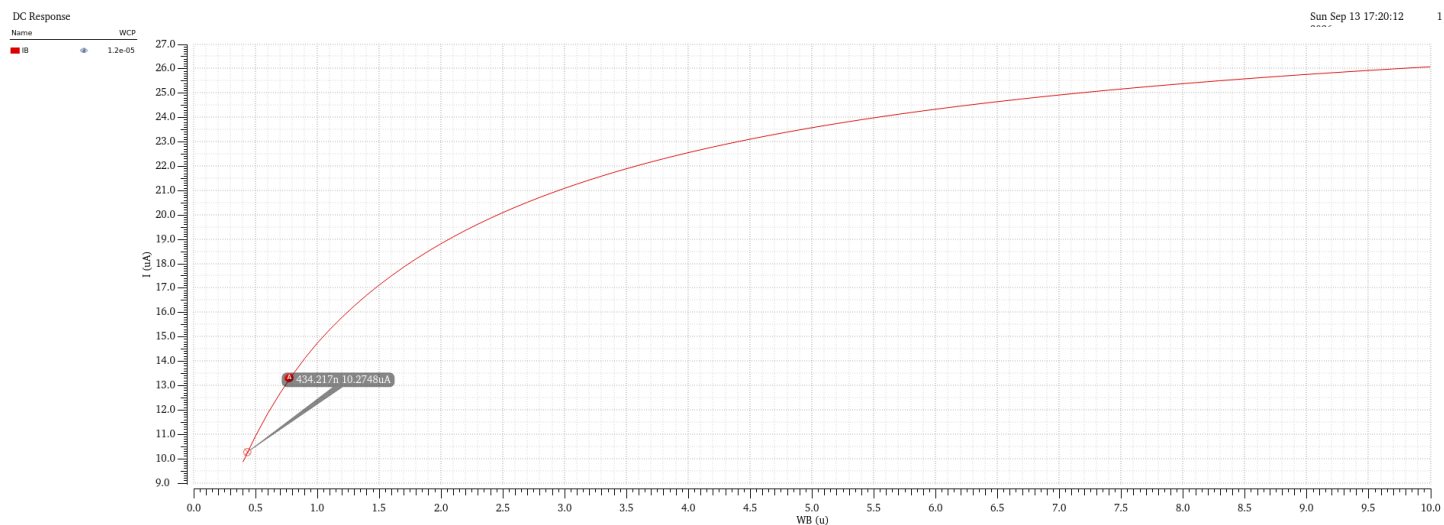
For the PMOS current source, we can select its width by sweeping it with the bias current in the current mirror transistor. Let $L_B = 1\mu m$



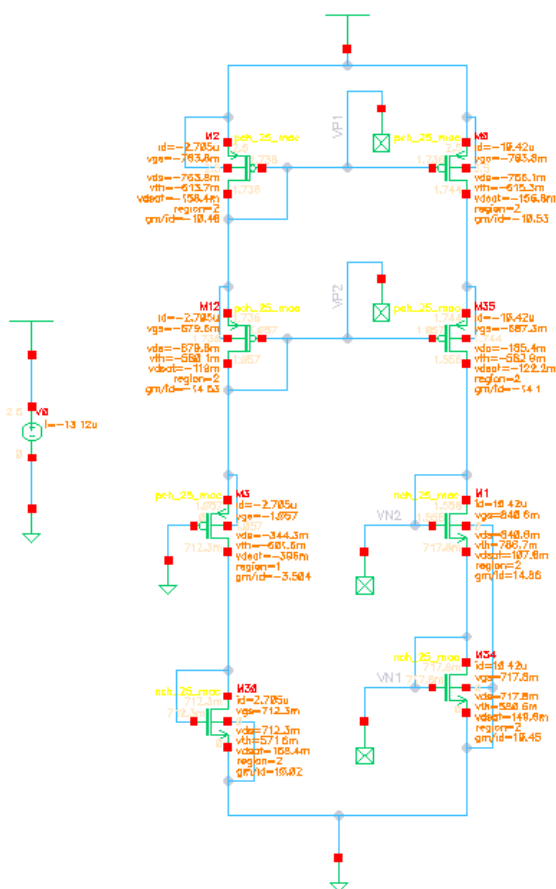
For better mismatch $\frac{\sigma_{I_{out}}}{I_{out}} = \sqrt{1 + \frac{1}{m}} \times \frac{I_{d_{mis}}}{I_D} \times 100\%$, used mirroring ratio $m = 4$

The founded results for the current mirrors mismatch on ADT:

$$\lambda_N = 80.3\text{mV}, \sigma_N = 3.78\%, \lambda_P = 45\text{mV}, \sigma_P = 1.54\%$$



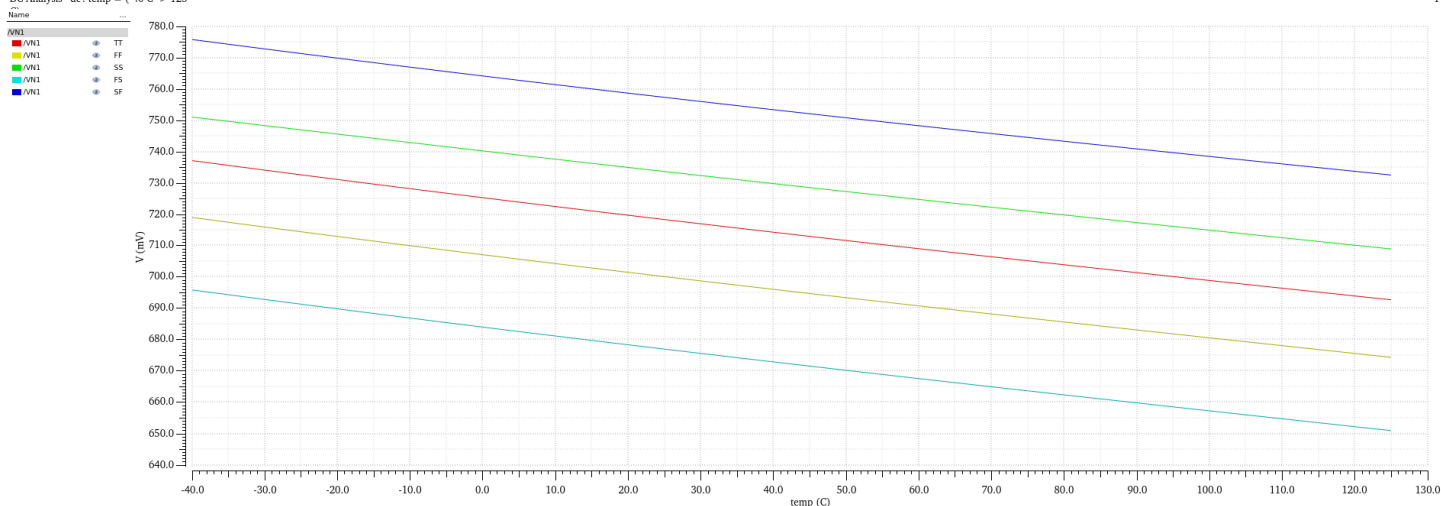
$$W_B = 0.45\mu\text{m}$$



Current mirror DC-OP simulation

DC Analysis 'dc': temp = (-40 C -> 125

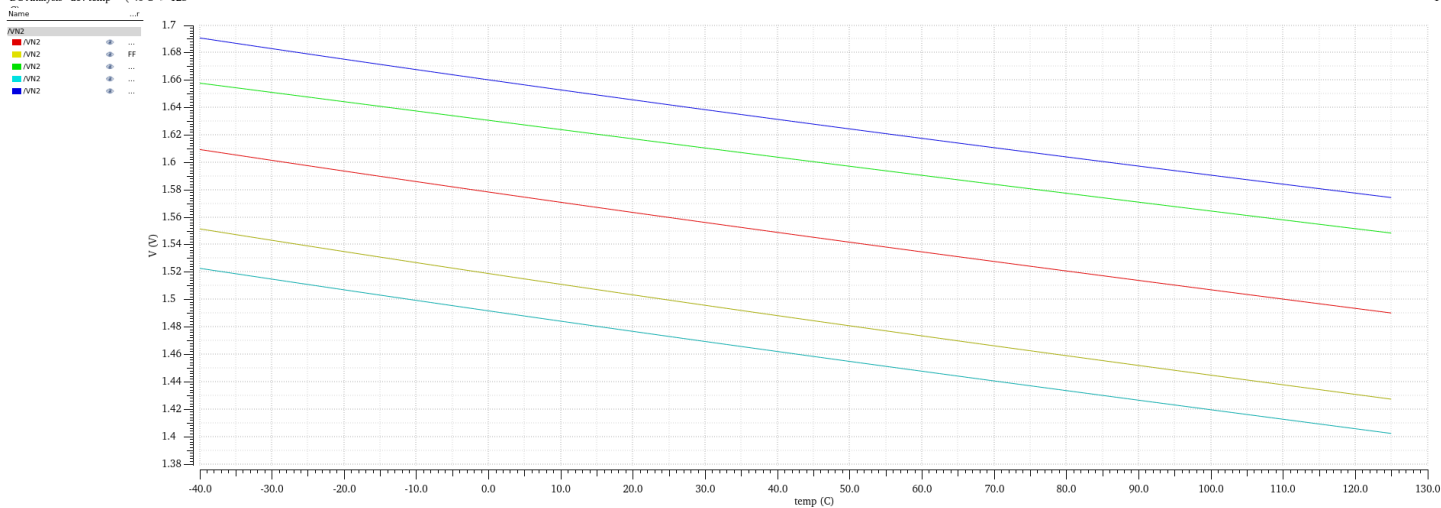
1



V_{N1} corners vs temperature

DC Analysis 'dc': temp = (-40 C -> 125

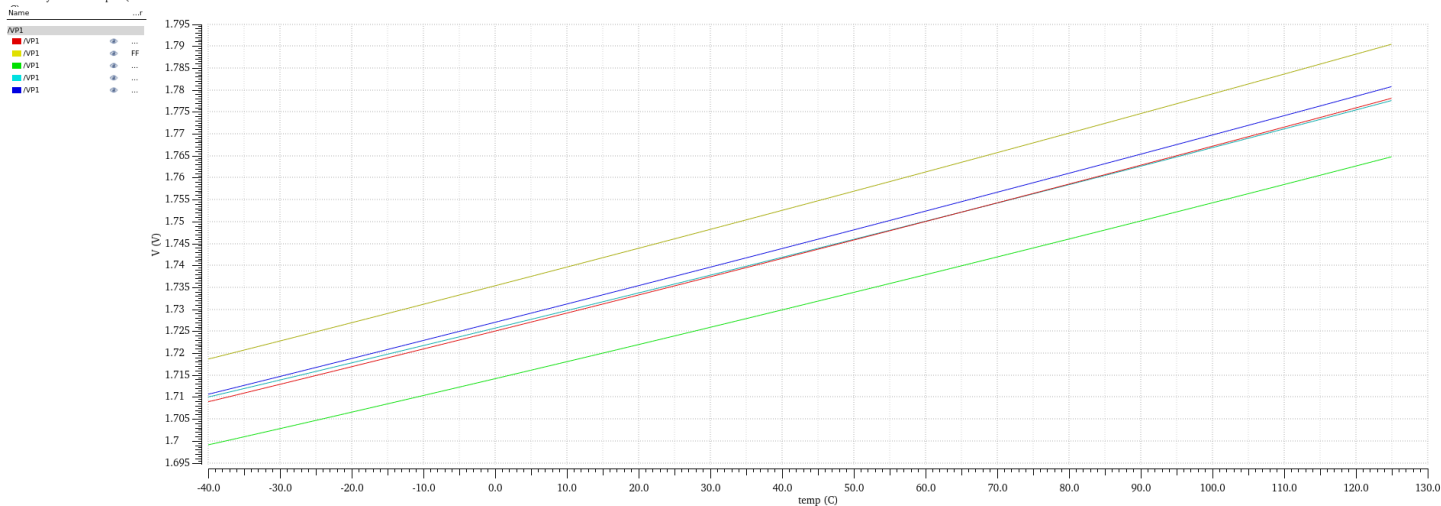
1



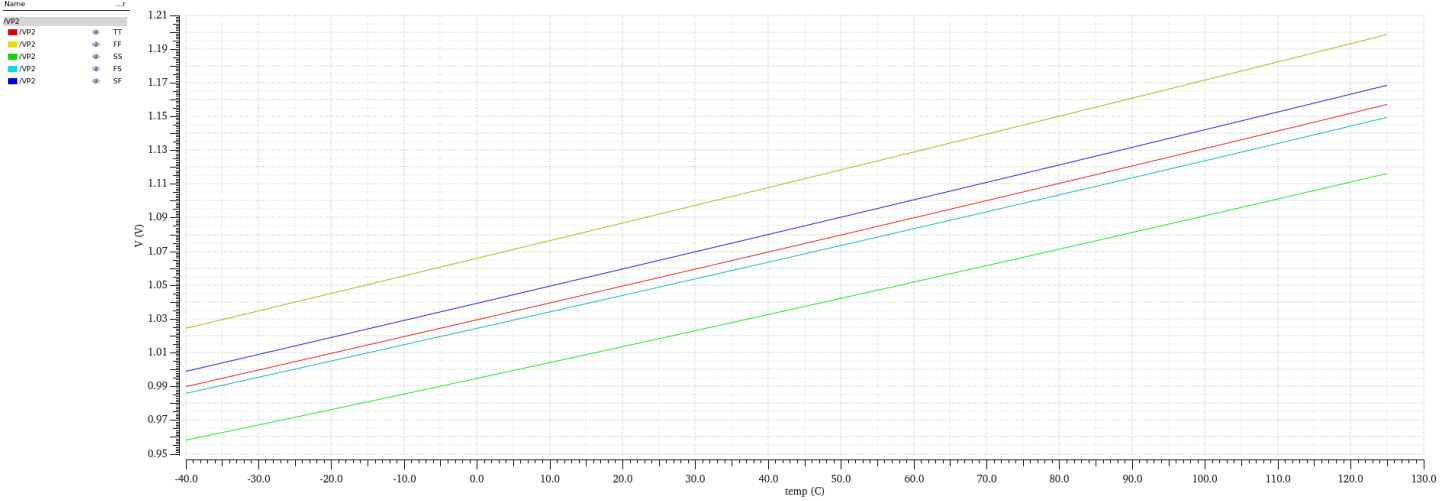
V_{N2} corners vs temperature

DC Analysis 'dc': temp = (-40 C -> 125

1



V_{P1} corners vs temperature

 V_{P2} corners vs temperature

	TT	FF	SS	FS	SF
V_{N1}	44.4mv	44.63mv	42.07mv	44.88mv	43.22mv
V_{N2}	119.2mv	124.2mv	109.3mv	120.4mv	116.4mv
V_{P1}	69.16mv	71.75mv	65.64mv	67.56mv	70.1mv
V_{P2}	167.1mv	174.1mv	157.9mv	163.5mv	169.5mv

The variations of bias nodes vs temperature on the range $-40^\circ\text{C} \rightarrow 125^\circ\text{C}$

Note width of the current source will need some tuning when it placed in the op-amp schematic, since in the above test bench, we don't consider the gate leakage that will face the current mirror in the op-amp circuit. I tuned it to be $2.8\mu\text{m}$.

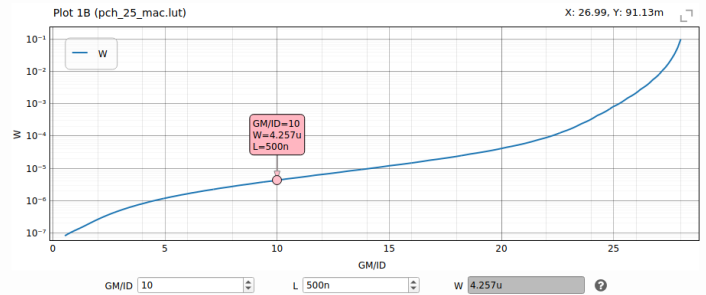
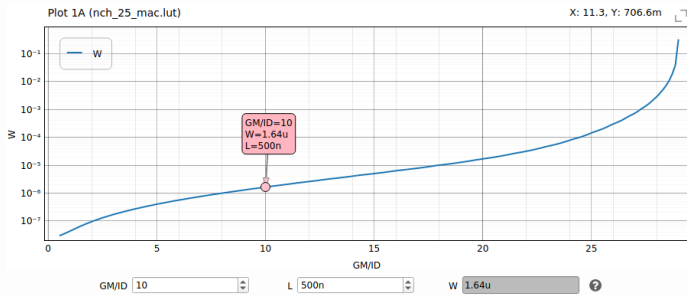
For the translinear loop, let $L = 0.5\mu\text{m}$, $g_m/I_D = 10$

$$\text{NMOS loop: } V_{GS24} + V_{GS25} = V_{GS28} + V_{GS16}$$

$$\text{PMOS loop: } V_{GS23} + V_{GS22} = V_{GS29} + V_{GS17}$$

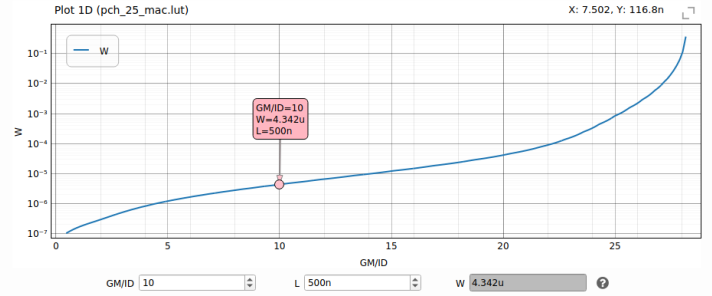
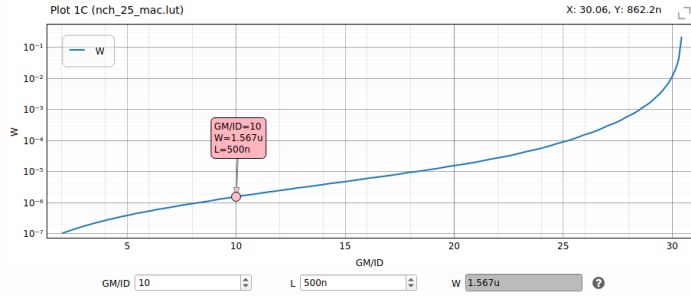
We need to match $M_{24} - M_{28}$, $M_{25} - M_{16}$, $M_{23} - M_{29}$, and $M_{22} - M_{17}$, so we can equate the width of each couple of them and set the multiplier value depending on each transistor drain current.

For $M_{24,28}$ and $M_{23,29}$, $V_{SB} = 0$, let $I_D = 10\mu\text{A}$, $V_{DS} = 1\text{V}$



$$W_{24,28} = 2\mu\text{m}, W_{23,29} = 5\mu\text{m} \text{ For } I_D = 10\mu\text{A}$$

For $M_{25,16}$ and $M_{22,17}$, let $I_D = 10\mu A$, $V_{DS} = 0.7V$, $V_{SB-25,16} = 1V$



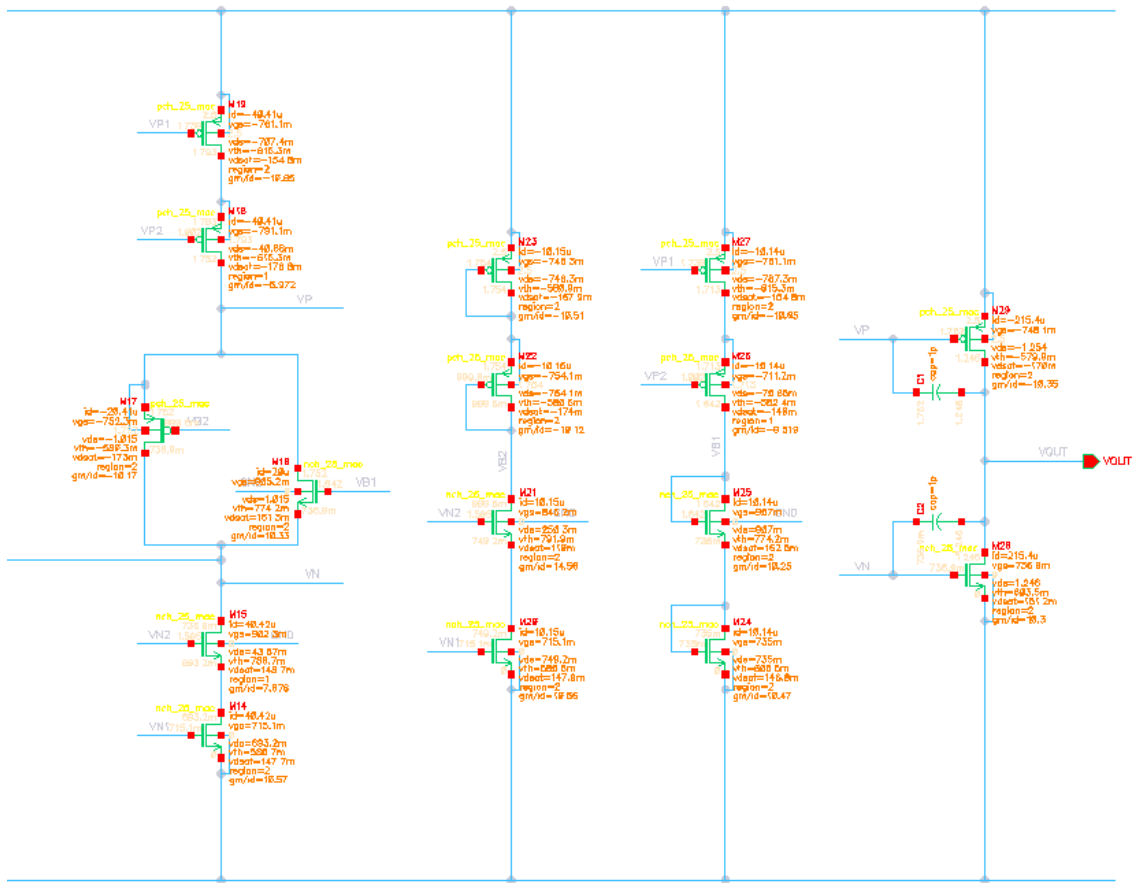
$$W_{25,16} = 1.8\mu m, W_{22,17} = 4.6\mu m \text{ For } I_D = 10\mu m$$

By taking the current in $M_{14,15}$ branch by $40\mu A$, the current in $M_{20,21}$ and $M_{24,25}$ branches by $10\mu A$, and the output stage current by $200\mu A$. We can set the width multiplier of each transistor depending on his current.

Let the compensation capacitors $C_M = 1pF$, we will tune them to meet the phase margin and GBW specs.

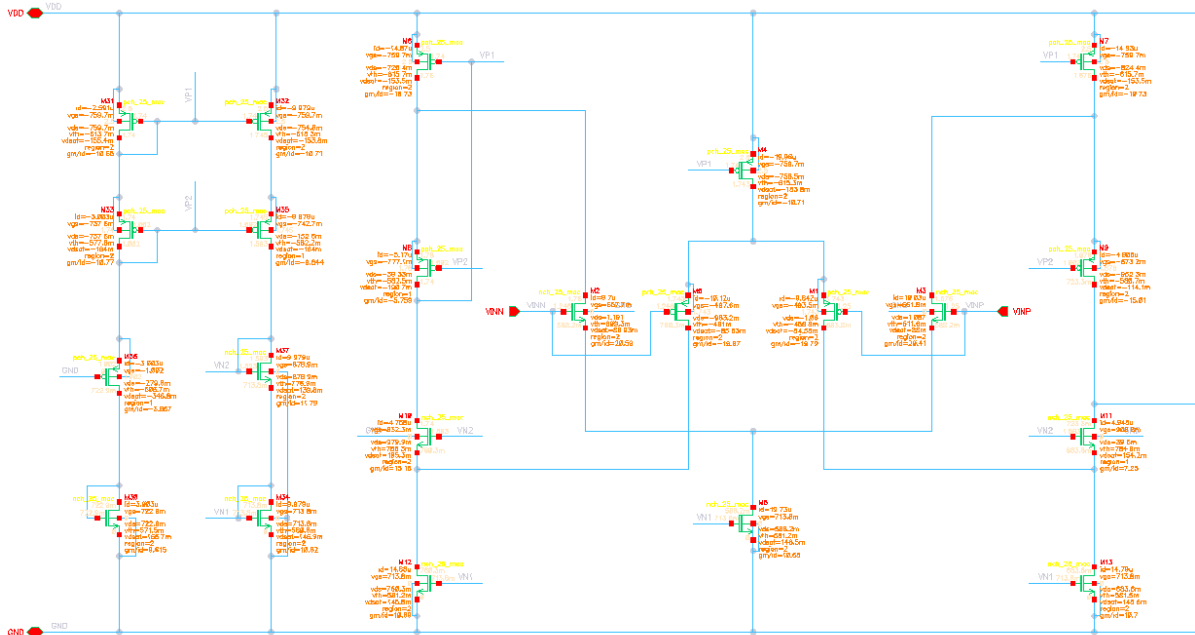
The total area is $367.95\mu m^2$ -

No load ($R_L = 1T\Omega$) OP: Current mirror and input stage

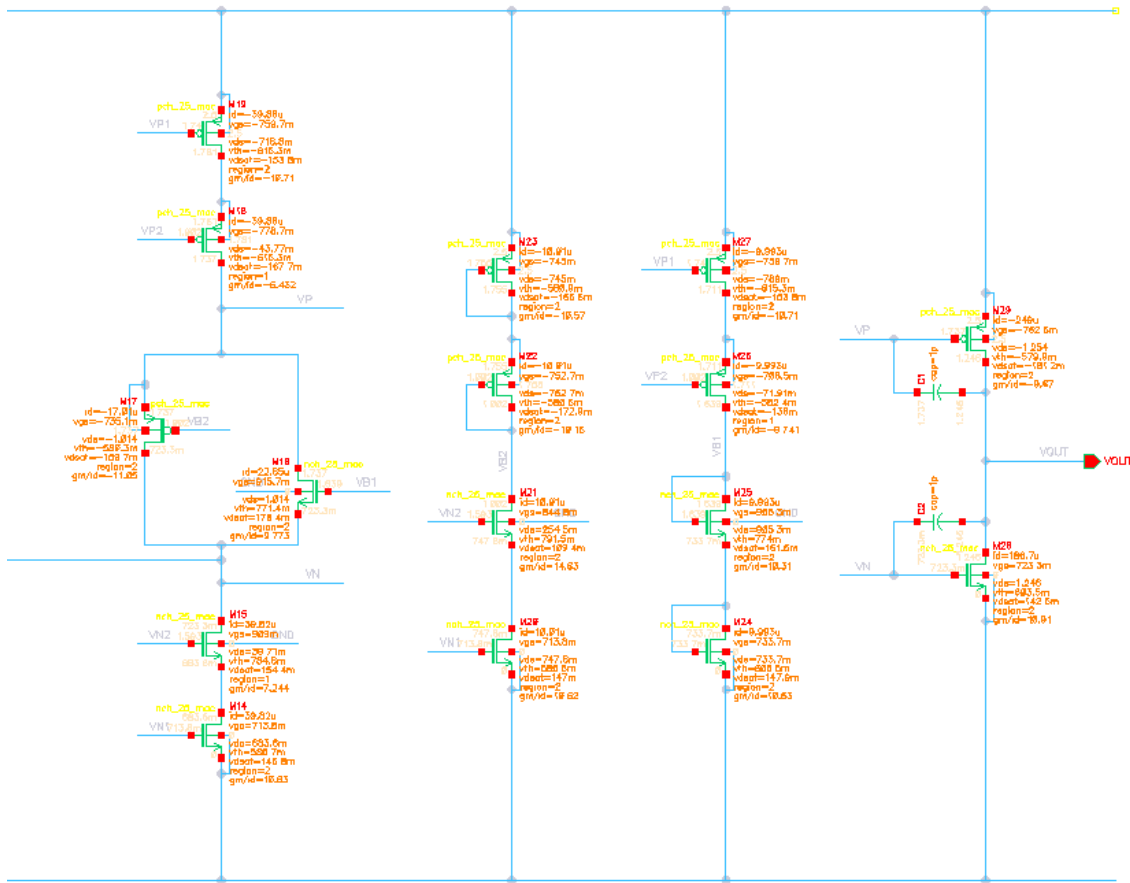


No load ($R_L = 17\Omega$) OP: Translinear loop and class AB output stage

Notice, you will find that each couple of $M_{28} - M_{24}$, $M_{29} - M_{23}$, $M_{25} - M_{16}$, & $M_{22} - M_{17}$ are well matched V_{GS}



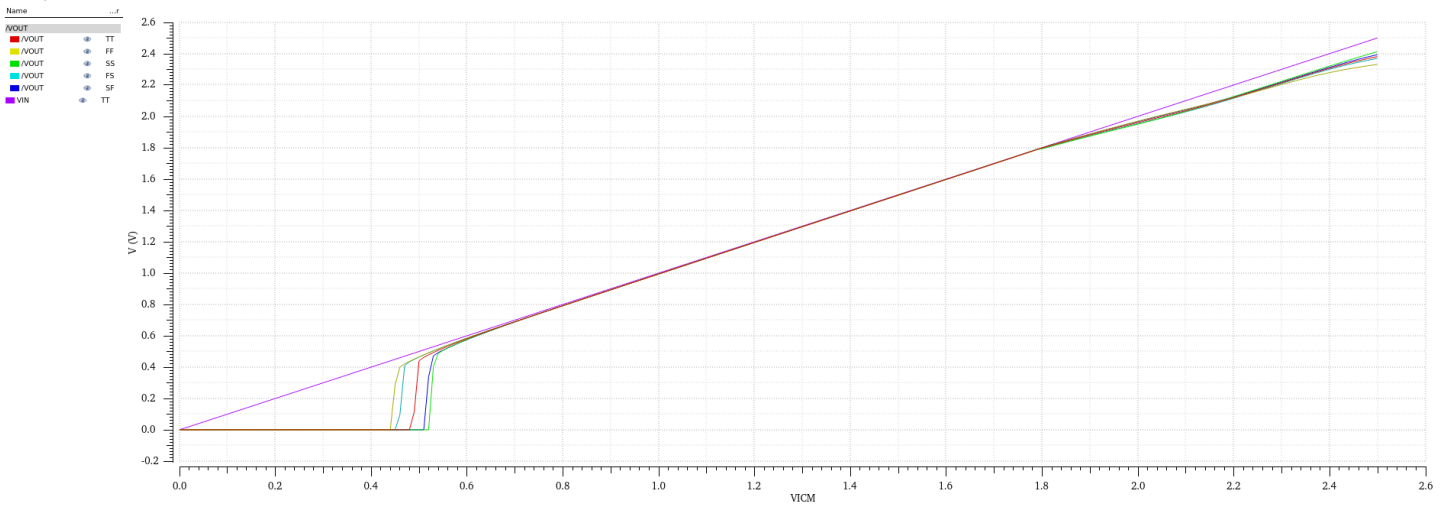
($R_L = 20k\Omega$) OP: Current mirror and input stage



($R_L = 20K\Omega$) OP: Translinear loop and class AB output stage

Notice, you will find that each couple of $M_{28} - M_{24}$, $M_{29} - M_{23}$, $M_{25} - M_{16}$, & $M_{22} - M_{17}$ are well matched V_{GS}

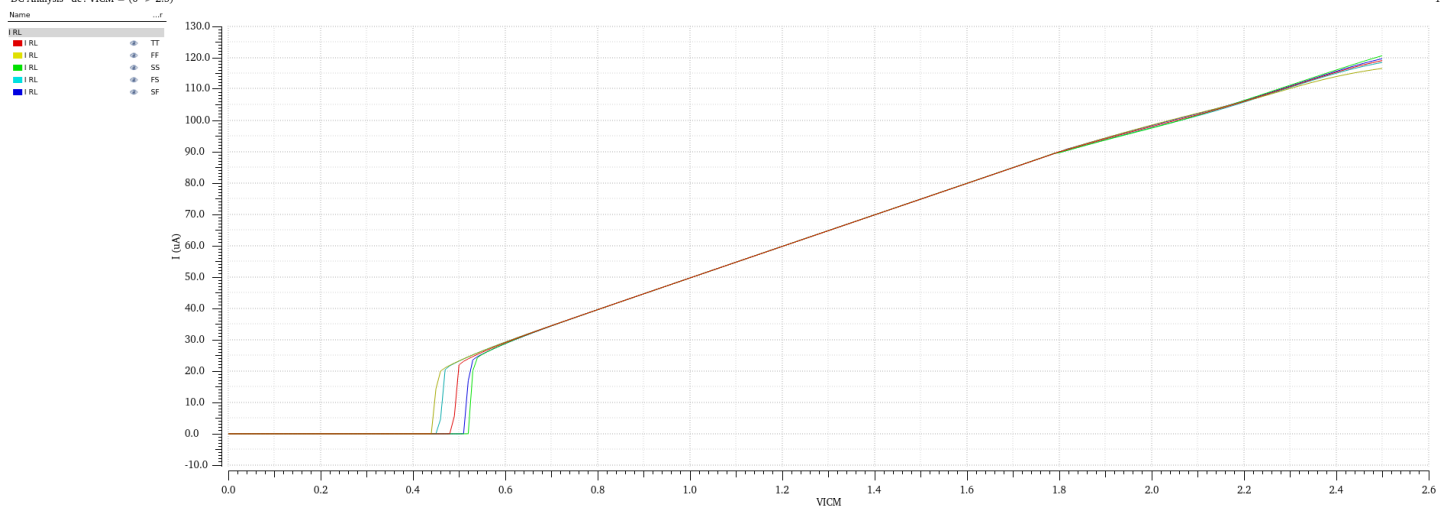
DC Analysis 'dc': VICM = (0 -> 2.5)



V_{out} vs $V_{ICM}(0 \rightarrow 2.5V)$

DC Analysis 'dc': VICM = (0 -> 2.5)

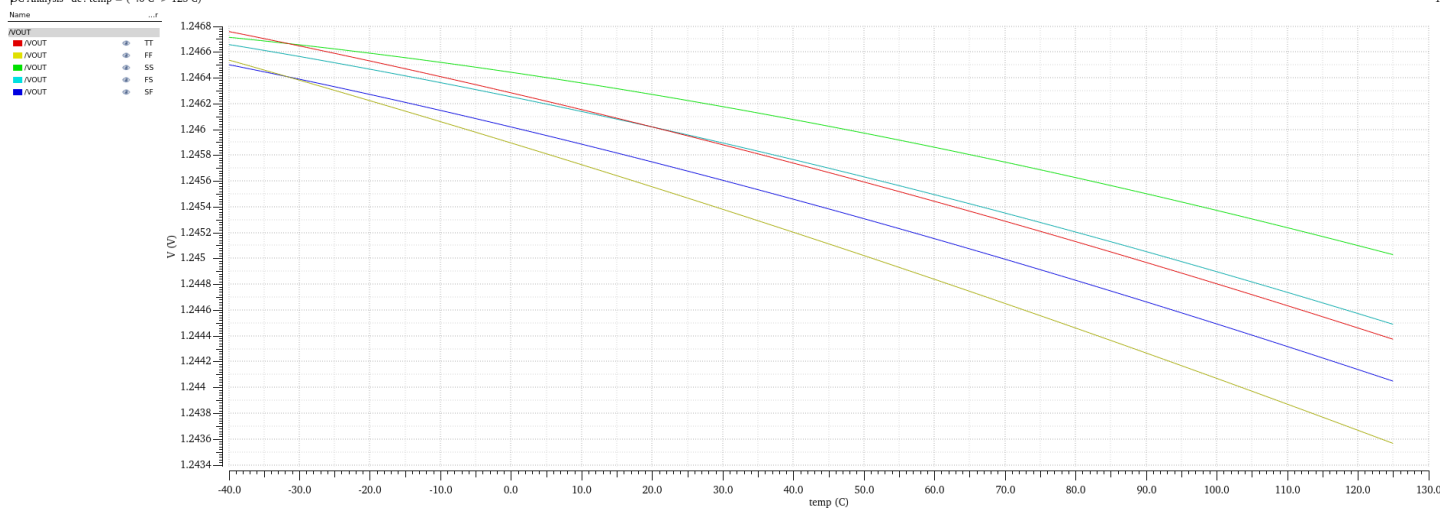
1



I_{RL} vs $V_{ICM}(0 \rightarrow 2.5V)$

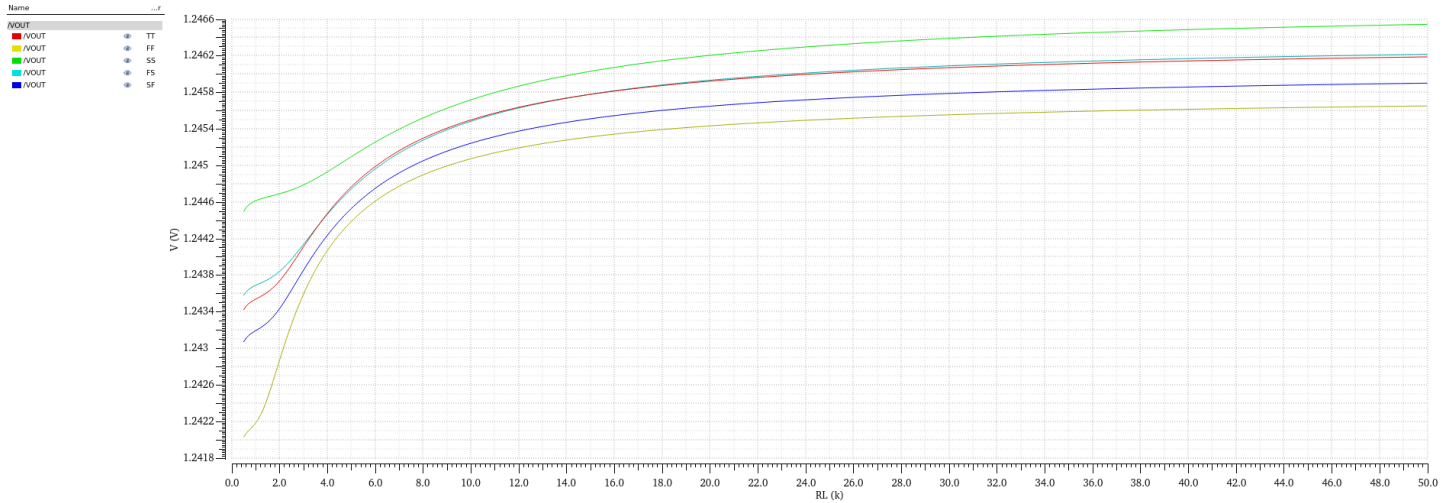
DC Analysis 'dc': temp = (-40 C -> 125 C)

1

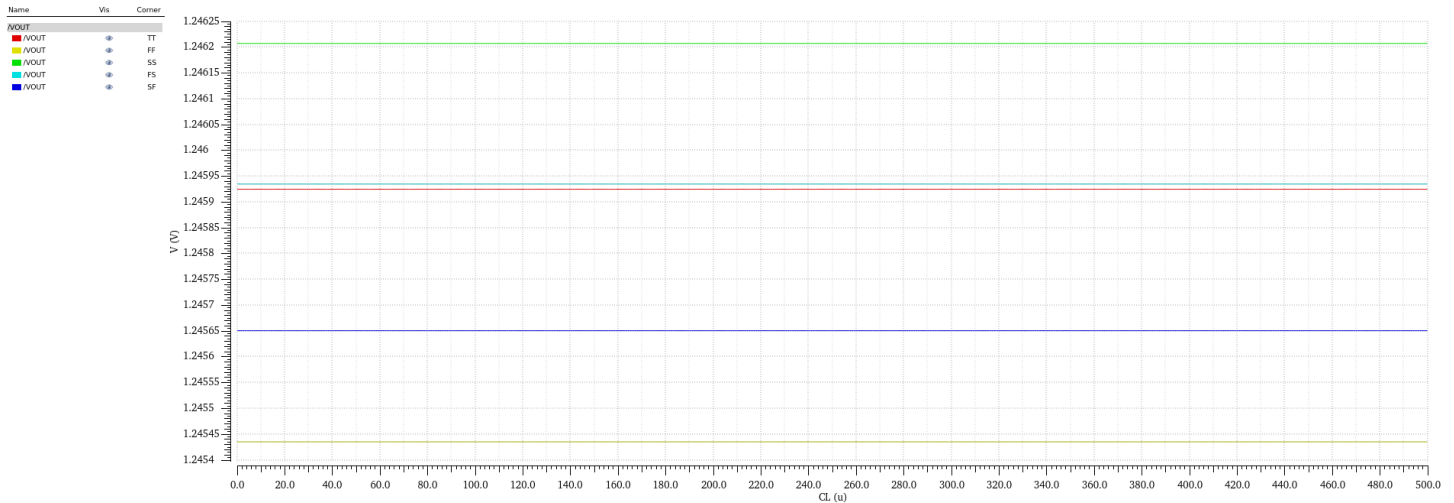


V_{out} vs temp(-40 → 125°C)

	TT	FF	SS	FS	SF
Voltage variations	2.385mV	2.97mV	1.684mV	2.167mV	2.451mV

 V_{out} vs $R_L(0.5 \rightarrow 50K\Omega)$

	TT	FF	SS	FS	SF
Voltage variations	2.765mV	3.62mV	2.045mV	2.634mV	2.828mV

 V_{out} vs $C_L(100fF \rightarrow 500\mu F)$

	TT	FF	SS	FS	SF
Voltage variations	5.757nV	1.381nV	888.2aV	350.2nV	29.6nV

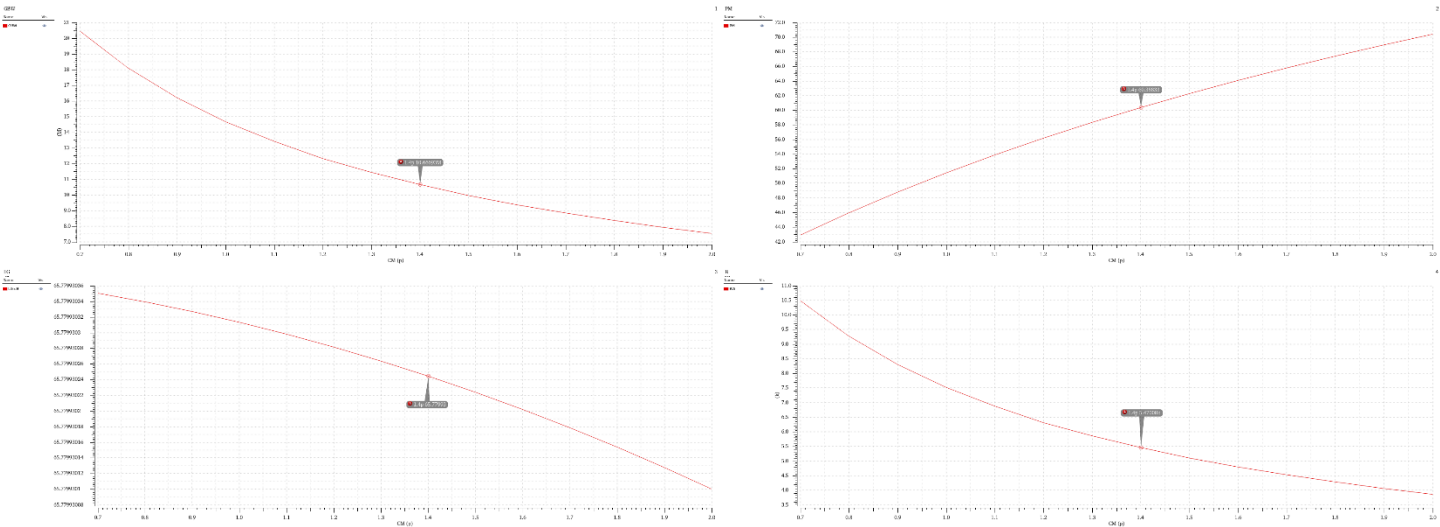
The output voltage variations with the capacitive load is almost negligible.

AC simulations

$LG \equiv A_{ol}$	$1.858K \equiv 65.38dB$
GBW	$14.74MHz$
UGF	$21.96MHz$
BW	$7.915KHz$
PM	47.83°
GM	$14.09dB$

AC results: $C_M = 1pF$

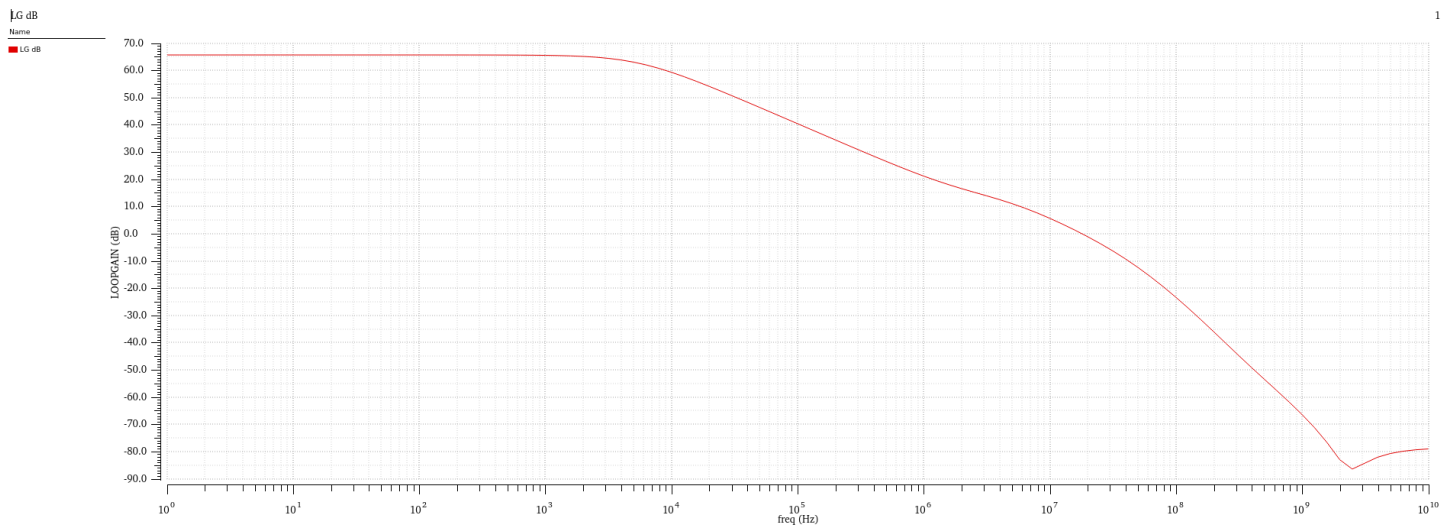
Now it is time to select the value of the compensation capacitors.



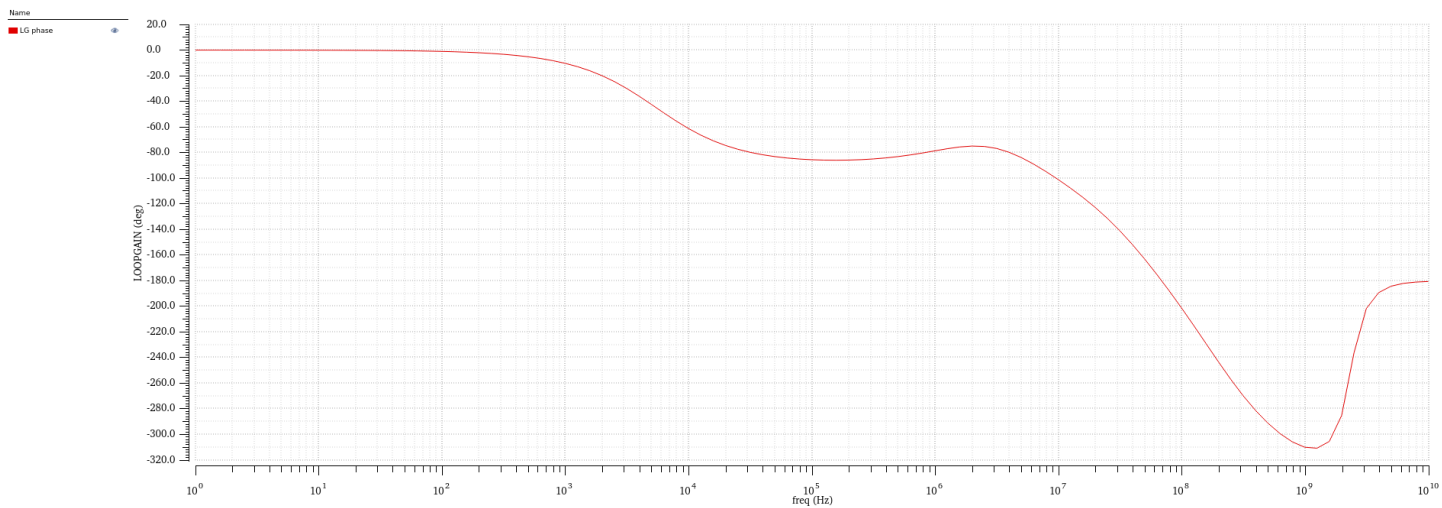
The value of C_M that can provide the required phase margin and GBW is $1.4pF$

$LG \equiv A_{ol}$	$1.945K \equiv 65.78dB$
GBW	$10.67MHz$
UGF	$18.44MHz$
BW	$5.47KHz$
PM	60.39°
GM	$17.09dB$

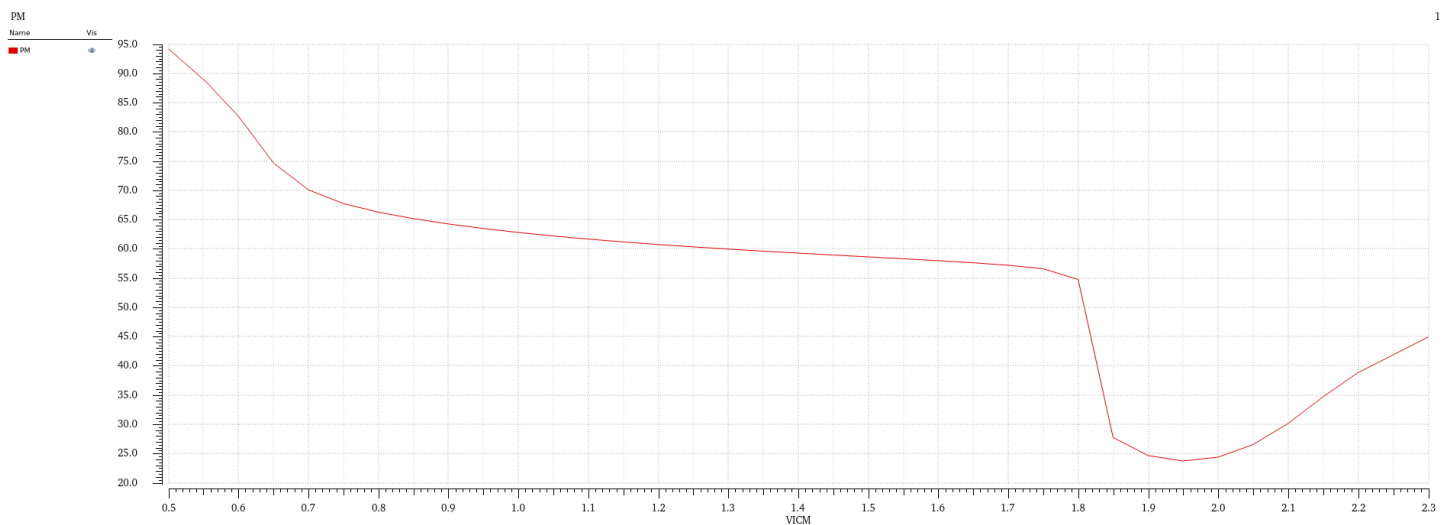
AC results: $C_M = 1.4pF$



Loop gain: $\beta = 1 \Rightarrow LG = A_{ol}$



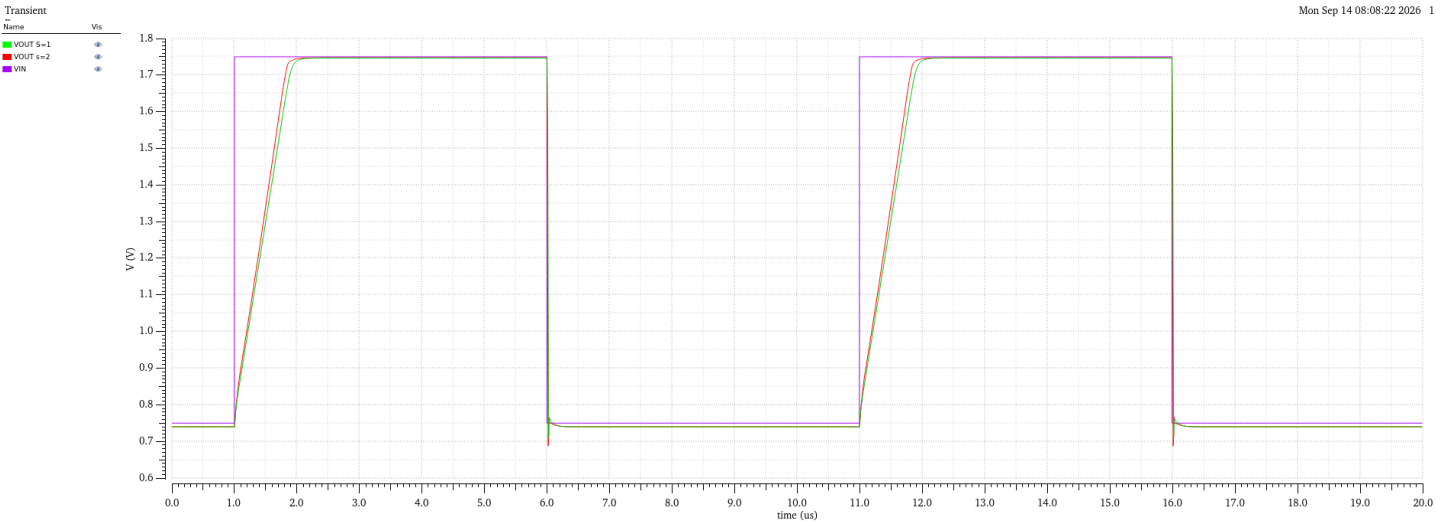
A_{ol} phase



PM vs V_{ICM}

Transient analysis

Applying 100KHz – 1V pk-to-pk pulse signal

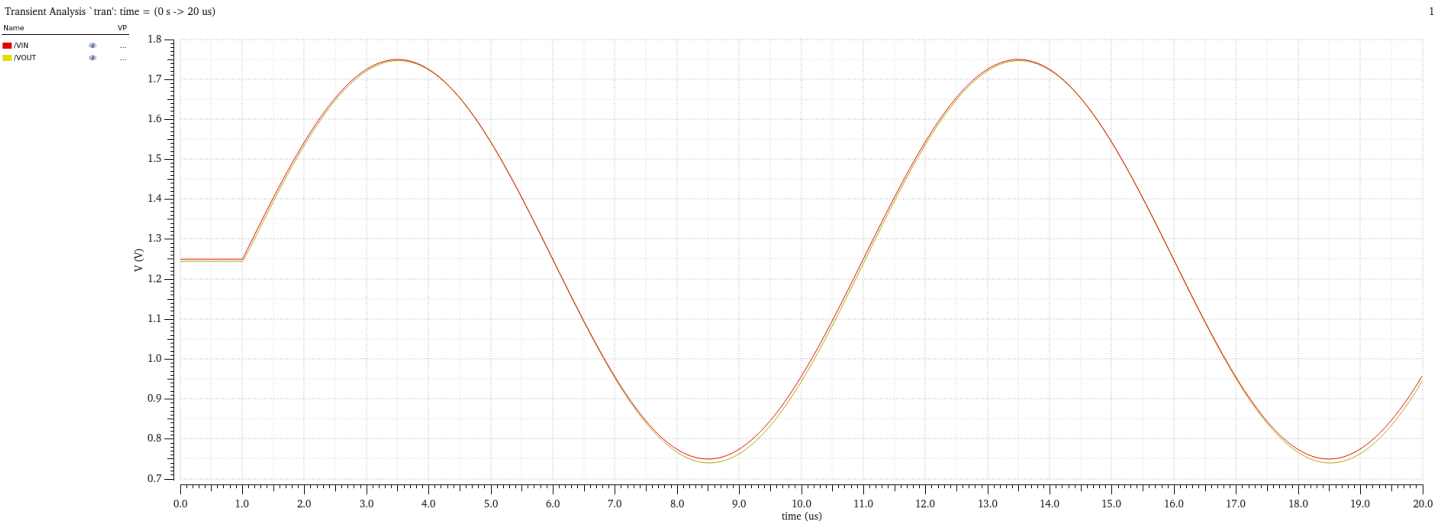


V_{out} for different current splitting ratio ■ $S = 1$ ■ $S = 2$

	$S = 1$	$S = 2$
$SR +$	1.03 MA/F	1.15 MA/F
$SR -$	40.7 MA/F	60.4 MA/F

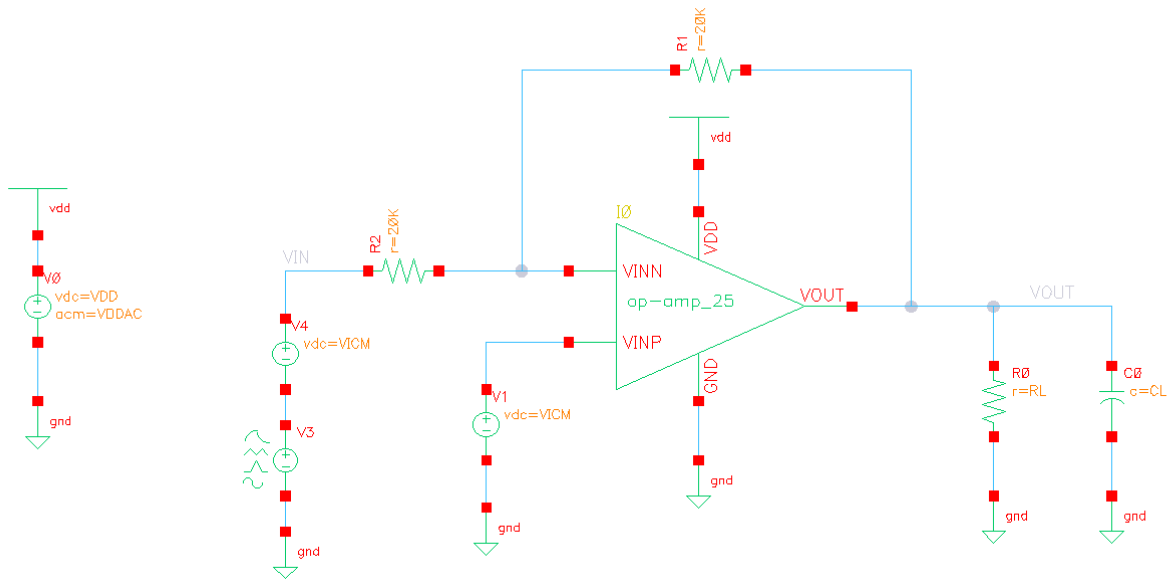
Not that, for $S = 2$ gives better transient results in the slew rate perspective as expected.

Applying 100KHz – 1V pk-to-pk sin signal



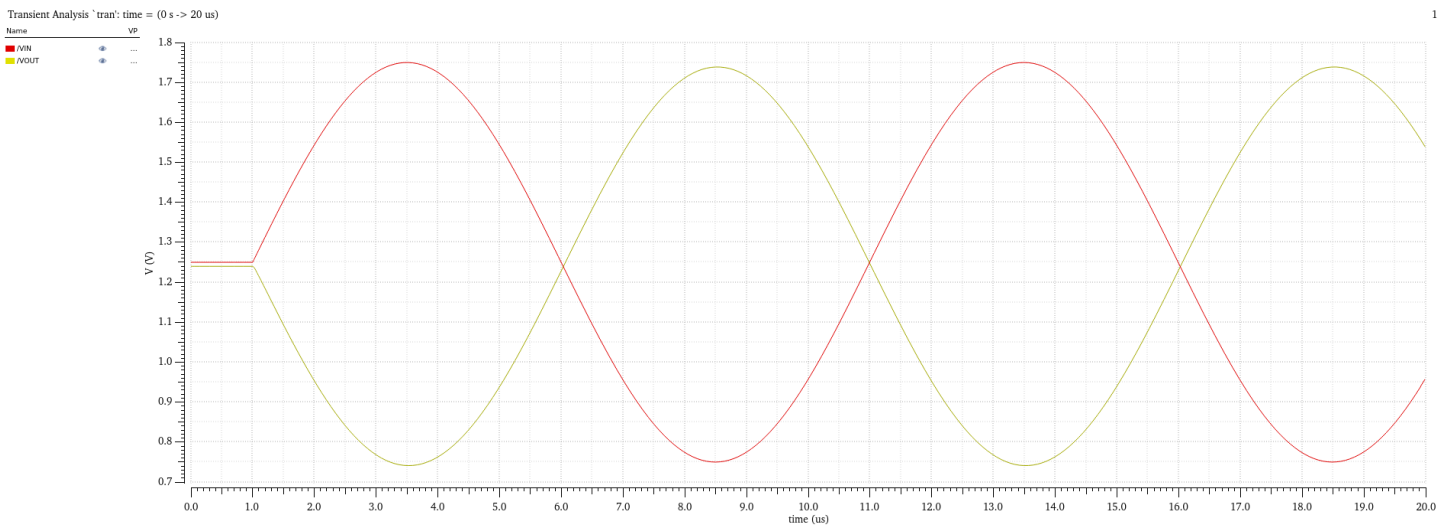
The output perfectly follows the input

Inverting amplifier topology



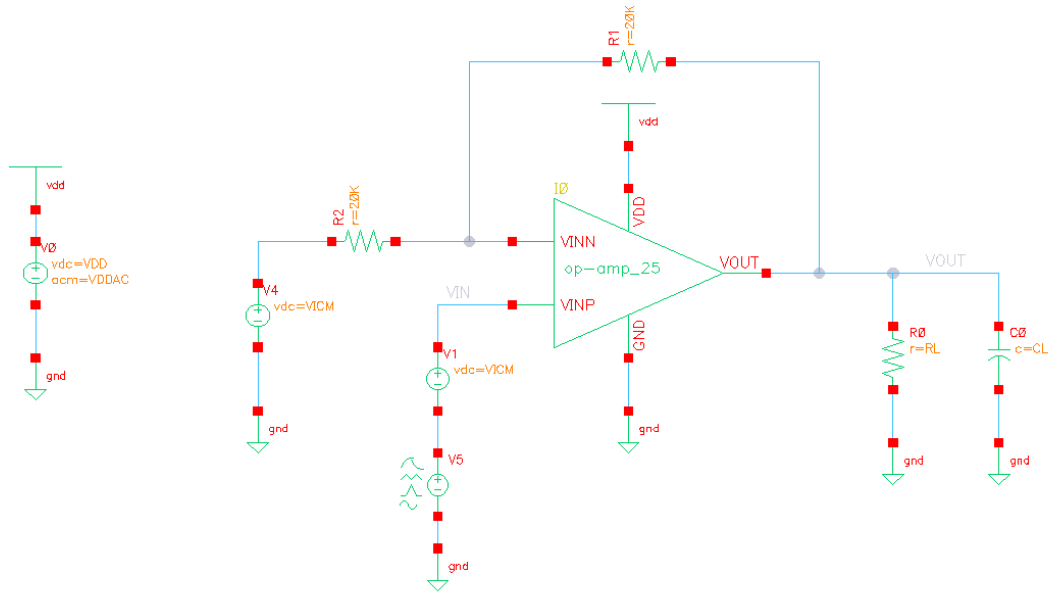
$$V_{out} \approx -\frac{R_1}{R_2} \times V_{in} = -V_{in} \Rightarrow 180^\circ \text{ phase shift with the same amplitude}$$

Applying 100KHz – 1V pk-to-pk sin signal



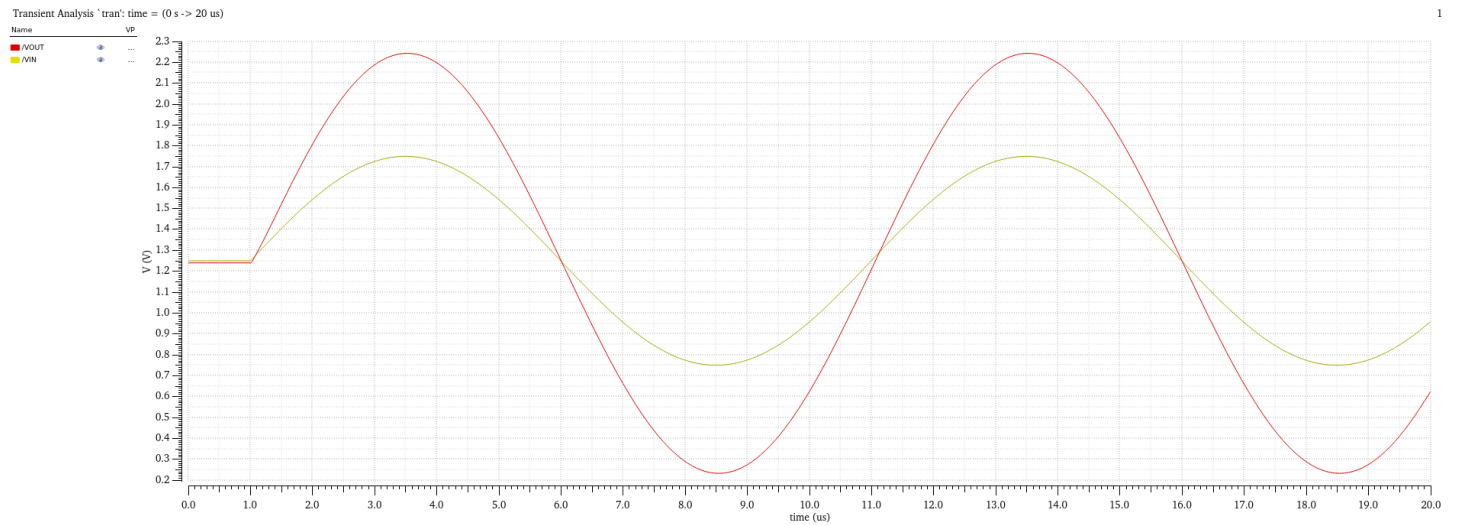
The output is a 180° shifted version of the input

Non-inverting amplifier topology



$V_{out} \approx \left(1 + \frac{R_1}{R_2}\right) \times V_{in} = 2V_{in} \Rightarrow$ the output in phase with the input with doubled amplitude.

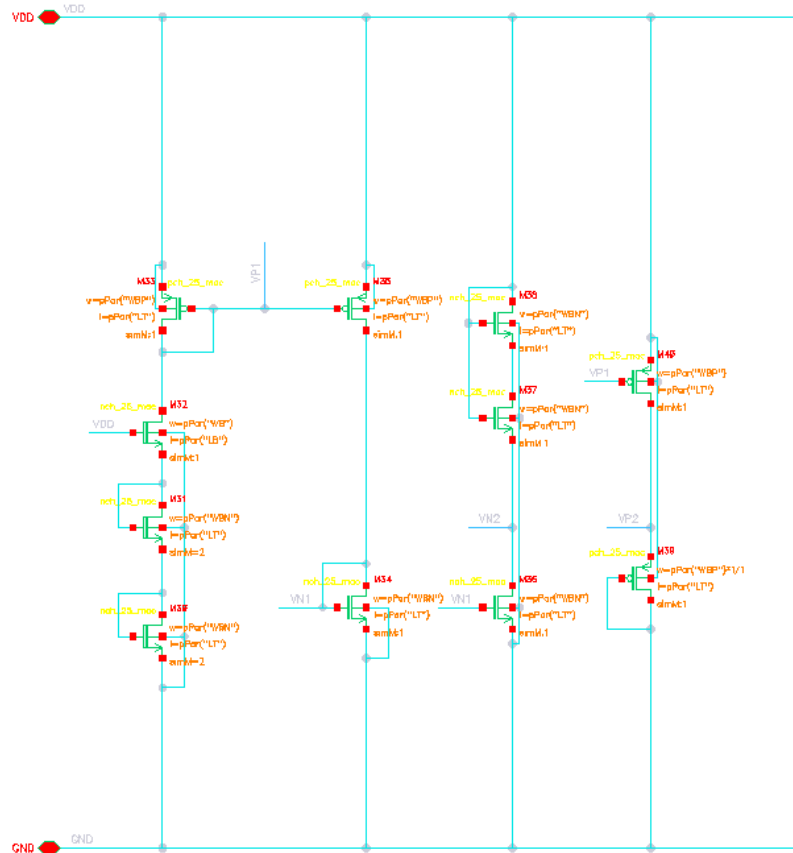
Applying 100KHz – 1V pk-to-pk sin signal



The output is doubled version of the input and in phase with it

Independent-Branch Cascode Bias Simulation

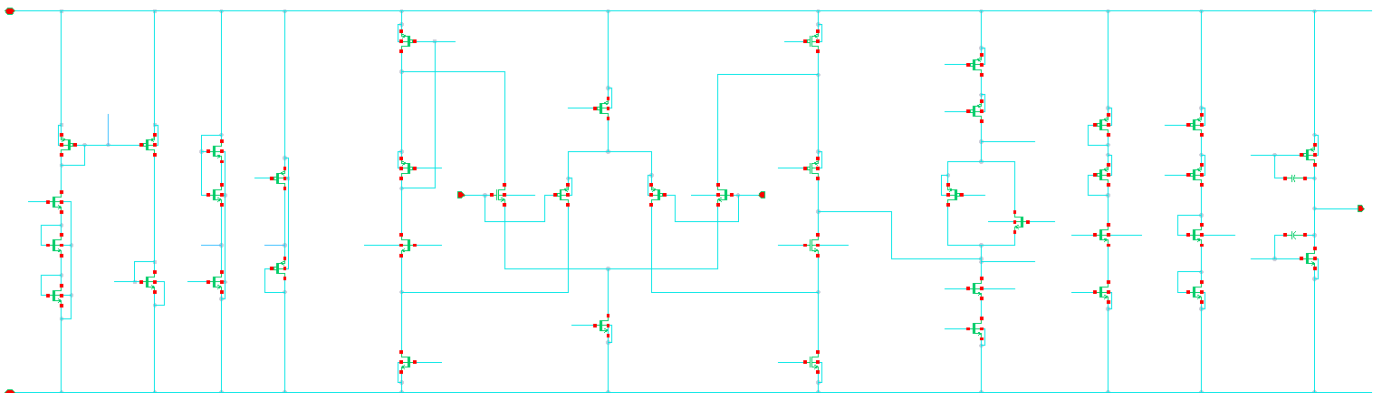
Instead of designing a single cascode current mirror for DC voltage biasing, we can make independent circuit for each node, that's would be area and power inefficient, but it can give better results.



This time we are using 2 independent branches to generate the bias voltage for the cascode transistors.

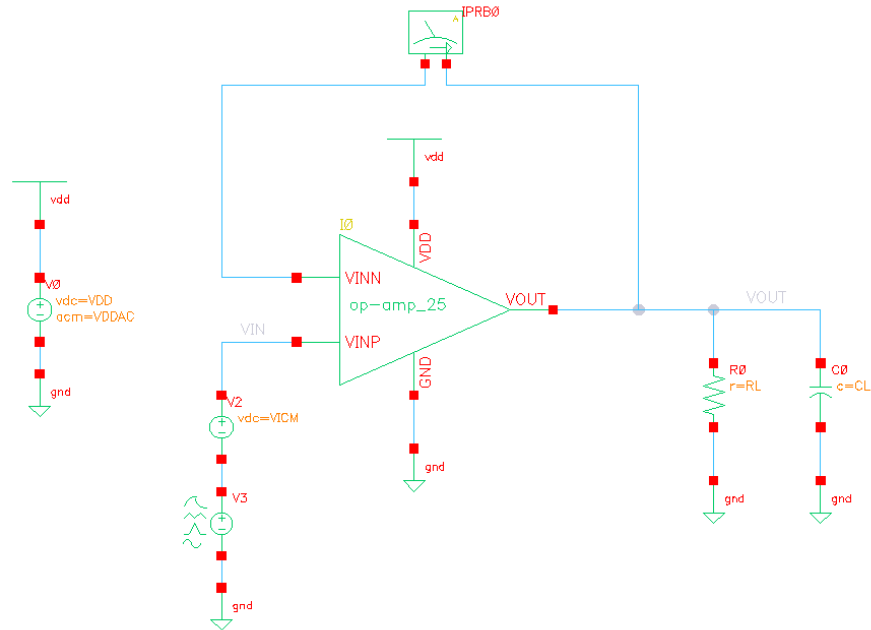
Designing the current mirror with mirroring ratio $m = 1$, by NMOS current source.

Using splitting ratio $S = 1$, the sizing is tuned a little bit.

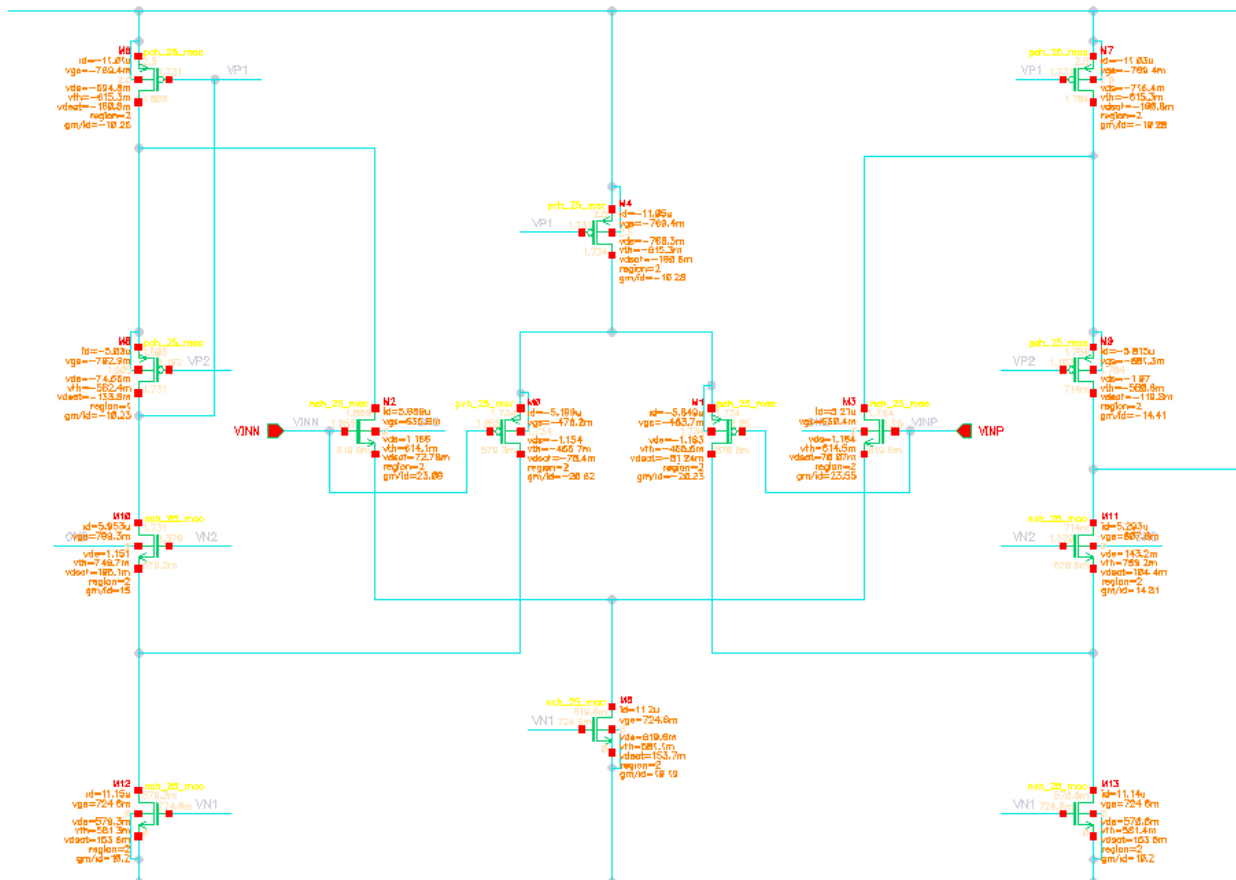


The input stage, translinear loop, and the class AB output stage are same as the 1st simulation

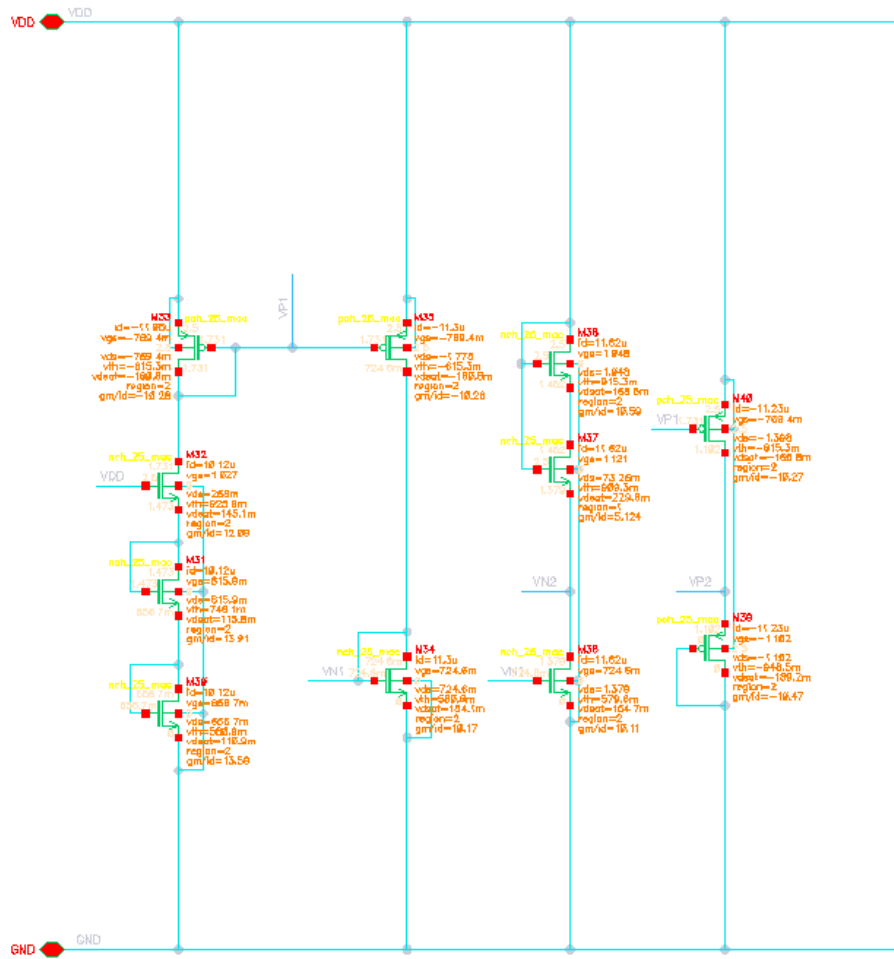
Buffer topology



DC OP simulation



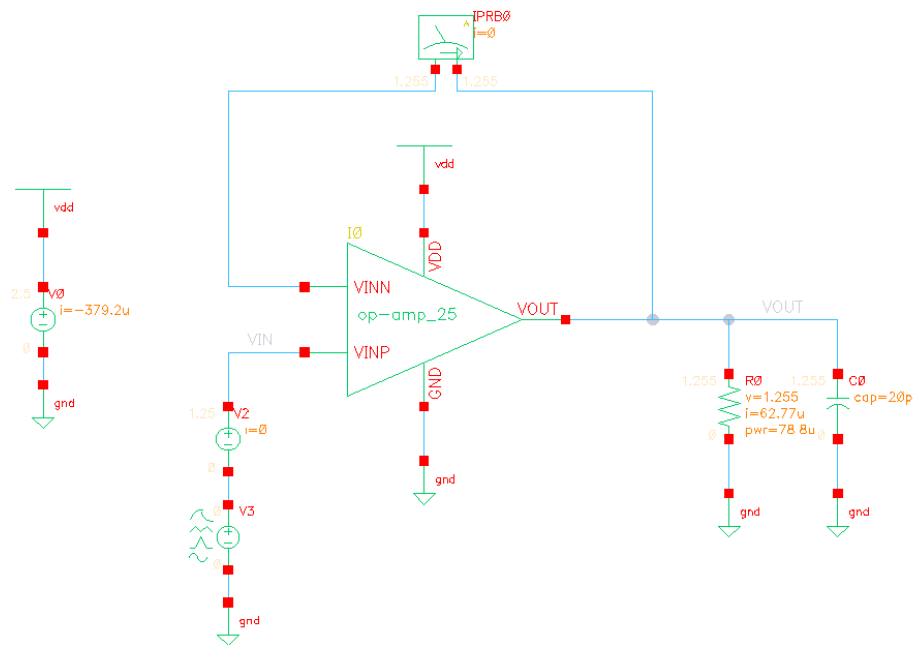
No load ($R_L = 1T\Omega$) OP: input stage



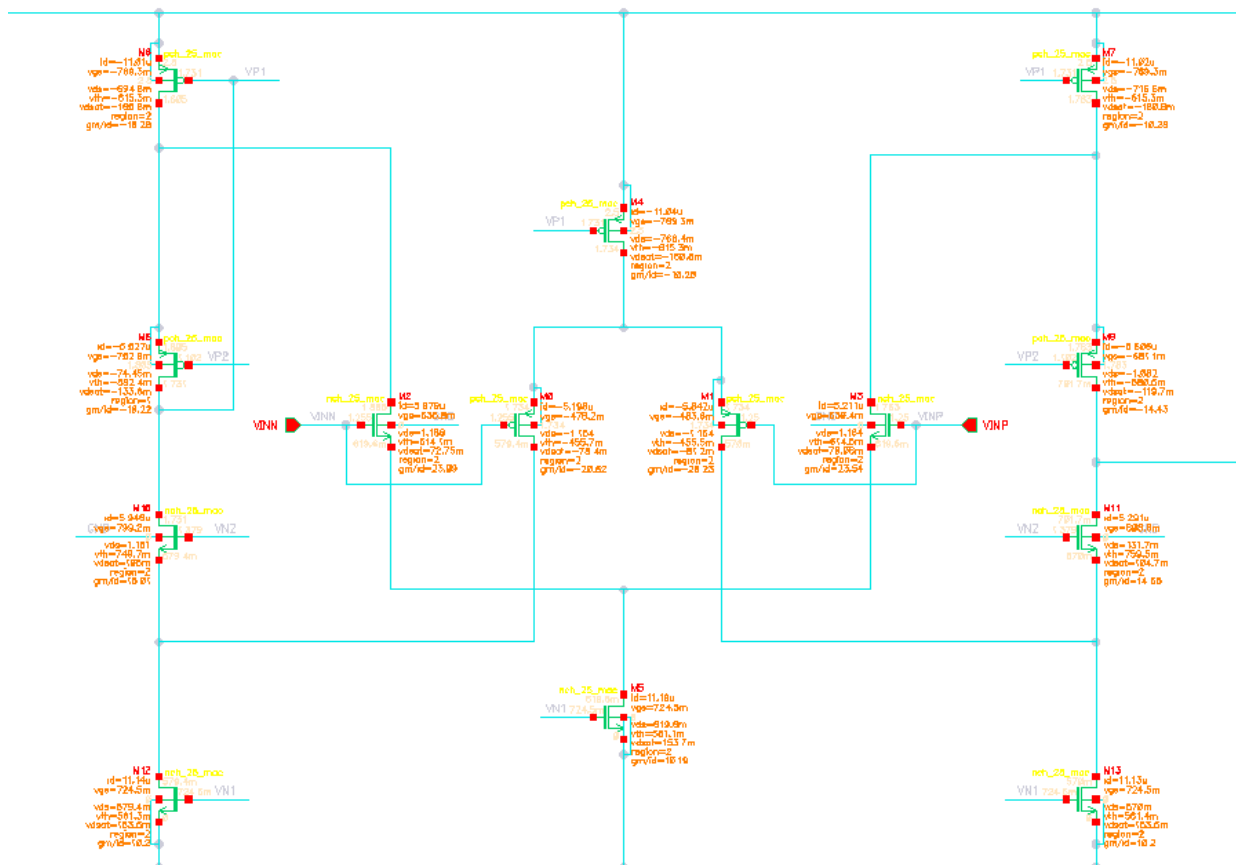
No load ($R_L = 17\Omega$) OP: Current mirror



No load ($R_L = 17\Omega$) OP: translinear loop and class AB output stage



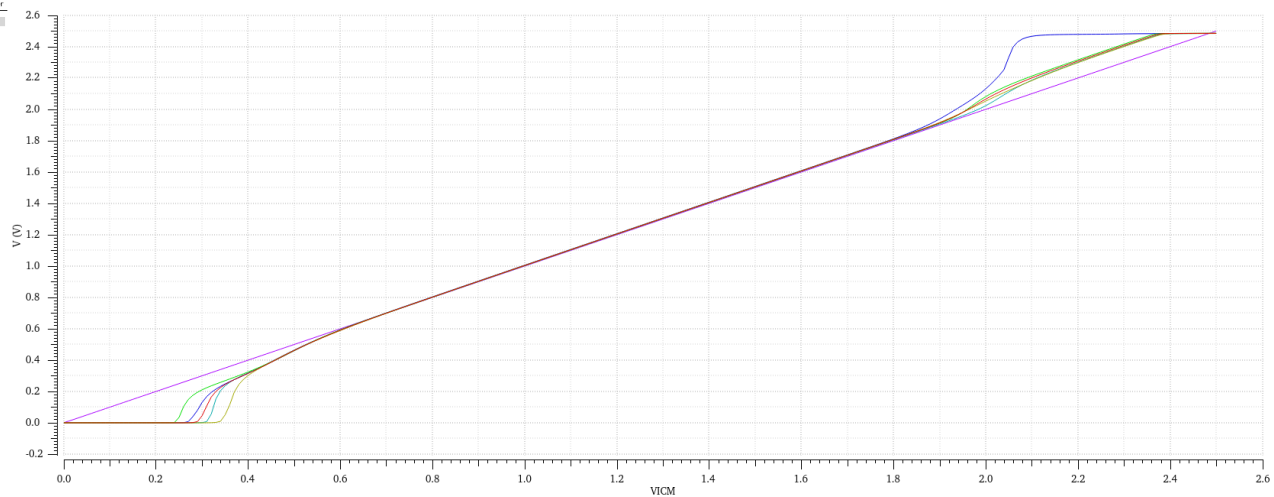
Testbench OP: ($R_L = 20K\Omega$)



($R_L = 20K\Omega$) OP: input stage

DC Analysis "VICM" = (0 ->

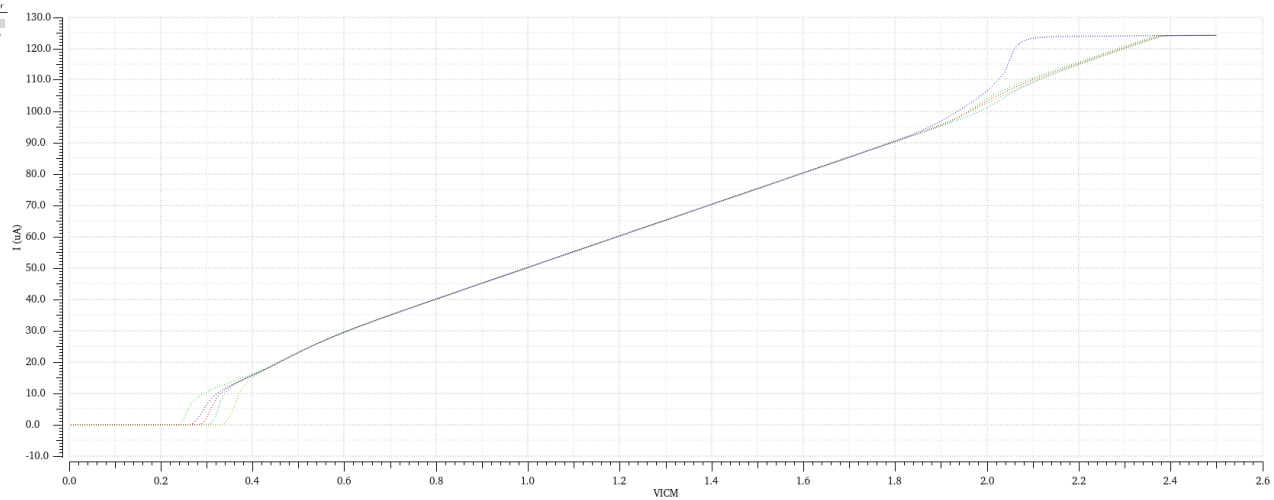
Name		
VOUT		nom
VOUT		FF
VOUT		SS
VOUT		SF
VOUT		FS
VIN		nom



V_{out} vs $V_{ICM}(0 \rightarrow 2.5V)$

DC Response

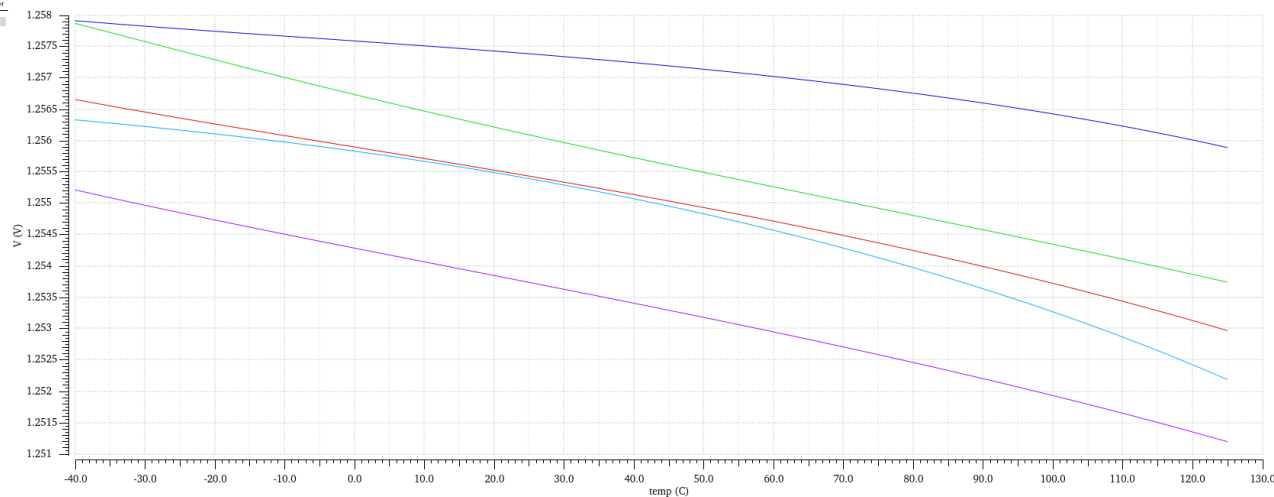
Name		
I RL		nom
I RL		FF
I RL		SS
I RL		SF
I RL		FS



I_{RL} vs $V_{ICM}(0 \rightarrow 2.5V)$

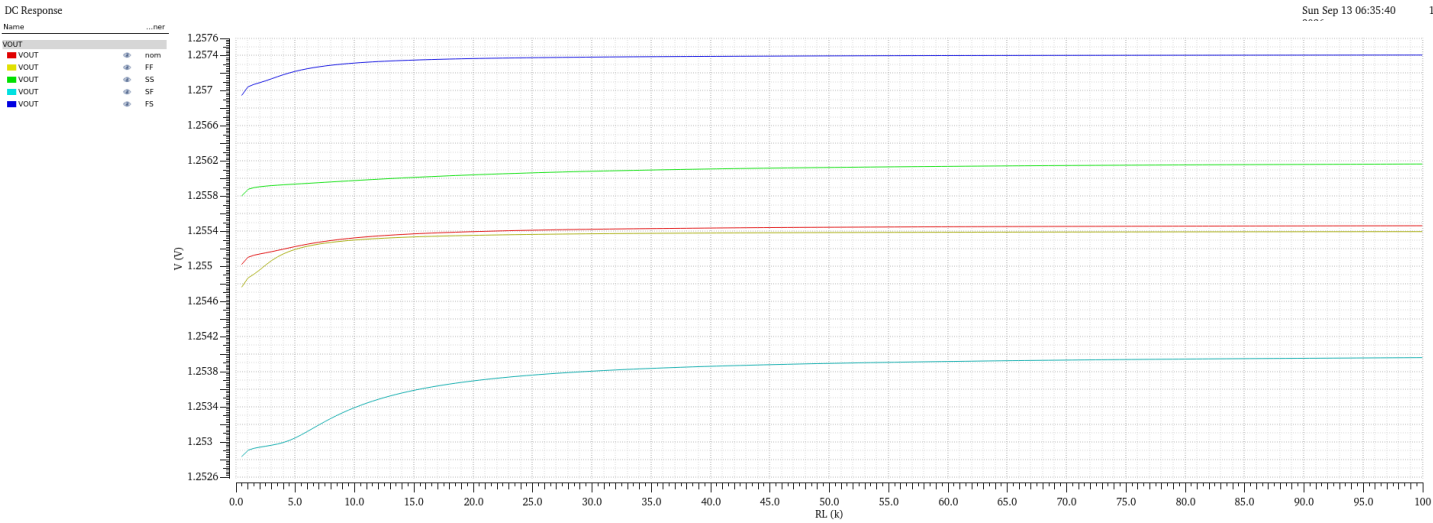
DC Analysis "temp" = (-40 C -> 125 C)

Name		
VOUT		nom
VOUT		FF
VOUT		SS
VOUT		SF
VOUT		FS

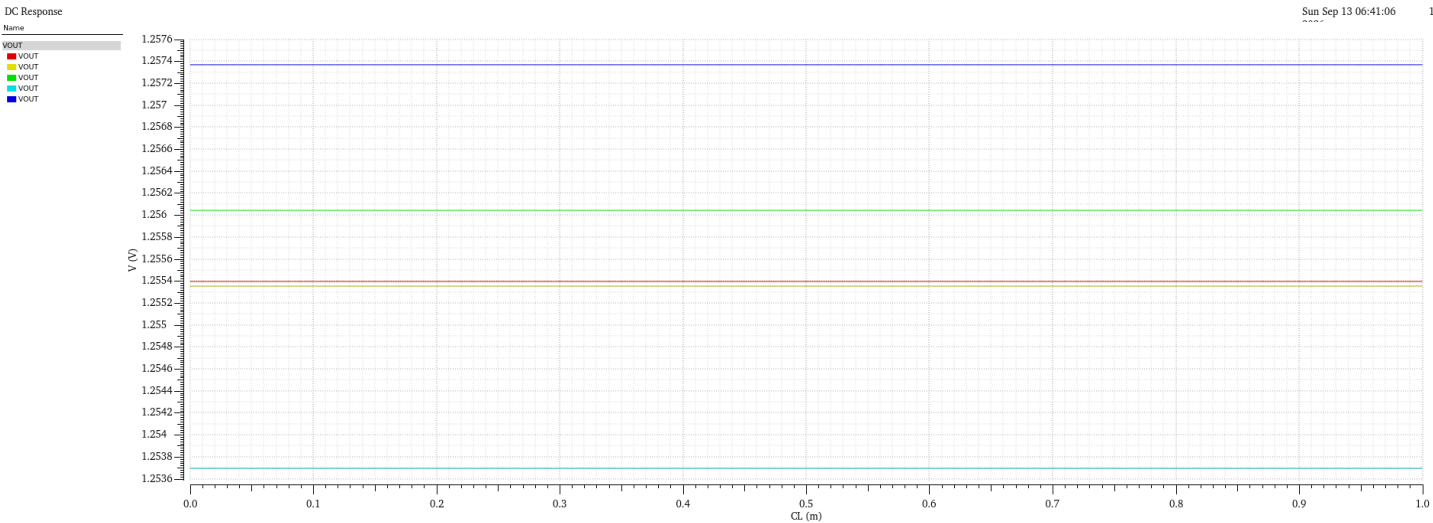


V_{out} vs temp(-40 $\rightarrow$ 125 $^{\circ}C$)

	TT	FF	SS	FS	SF
Voltage variations	3.685mV	4.143mV	4.129mV	2.024mV	4.015mV



	TT	FF	SS	FS	SF
Voltage variations	437.2 μ V	628.8 μ V	355.8 μ V	456 μ V	1.12mV

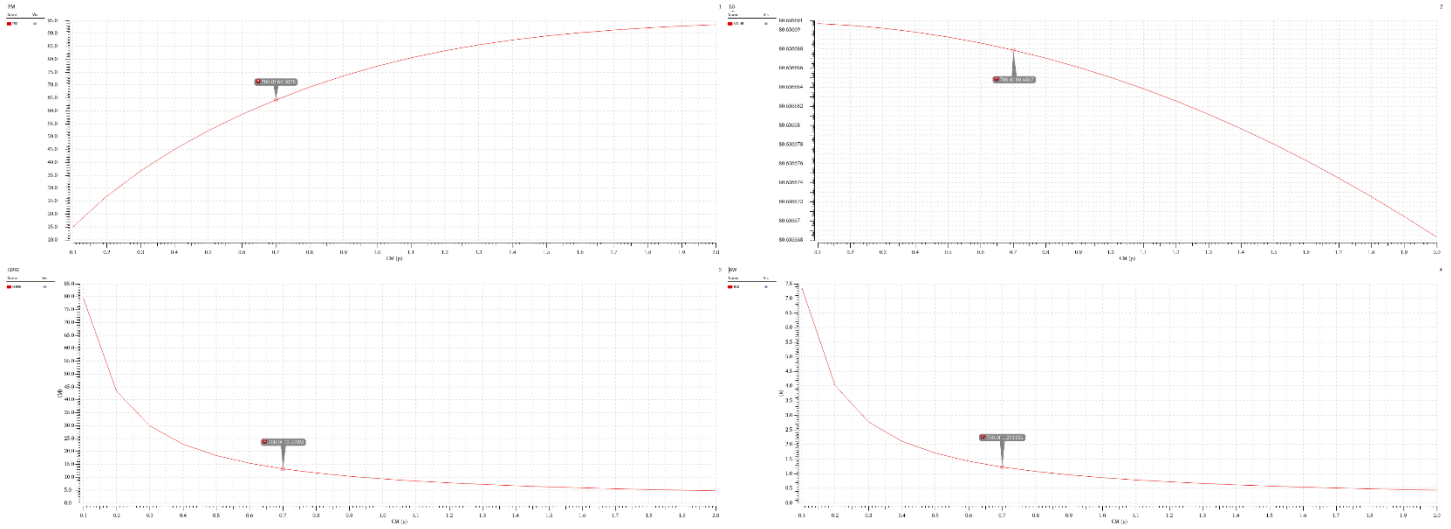


	TT	FF	SS	FS	SF
Voltage variations	4.213nV	1.648 μ V	167.1nV	2.205pV	12.76nV

The output voltage variations with the capacitive load is almost negligible.

AC simulations

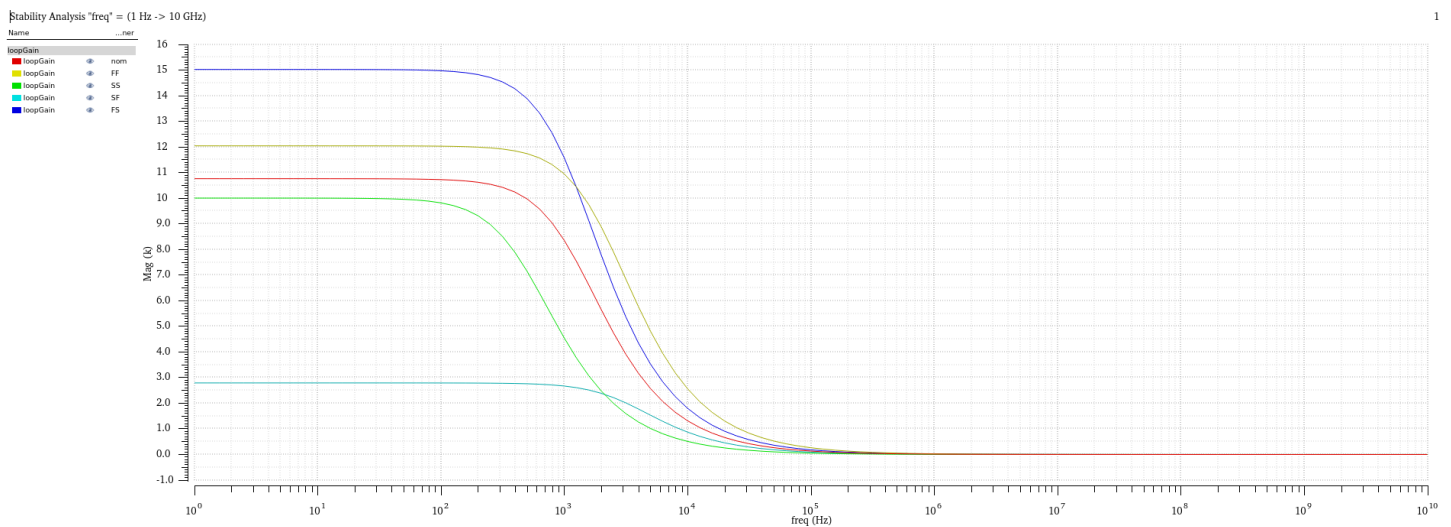
Now it is time to select the value of the compensation capacitors.



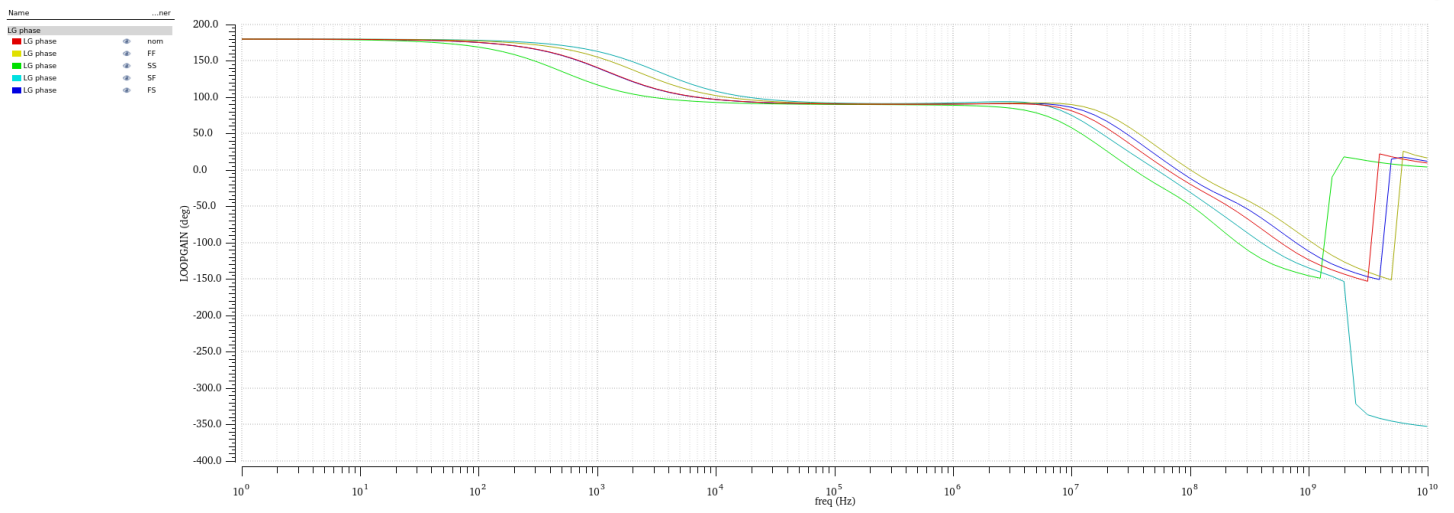
The value of C_M that can provide the required phase margin and GBW is $0.7pF$

	TT	FF	SS	FS	SF
$LG \equiv A_{ol}$	80.64 dB	81.61 dB	80 dB	83.57 dB	68.91dB
GBW	13.28MHz	26.38MHz	5.113MHz	18.21MHz	9.212MHz
UGF	17.18MHz	34.34MHz	5.804MHz	23.73MHz	13.01MHz
BW	1.231KHz	2.186KHz	0.51KHz	1.209KHz	3.293KHz
PM	64.31°	54.1°	75.98°	59.85°	65.1°
GM	18.23 dB	15.93 dB	21.63 dB	16.87 dB	19.17 dB

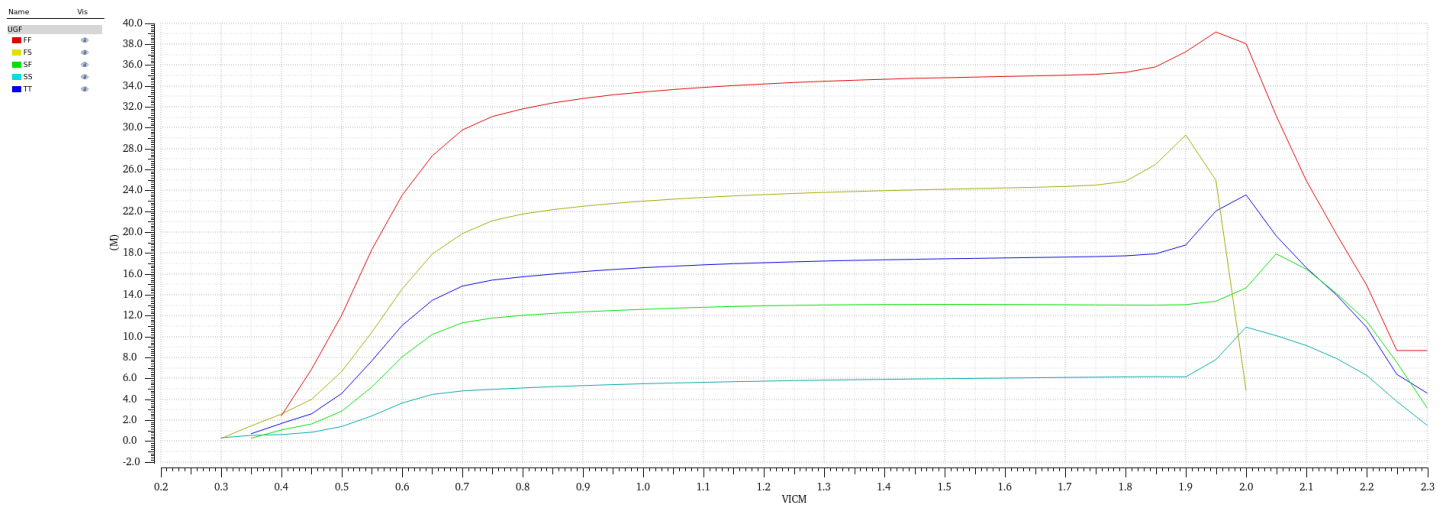
AC results: $C_M = 0.7pF$



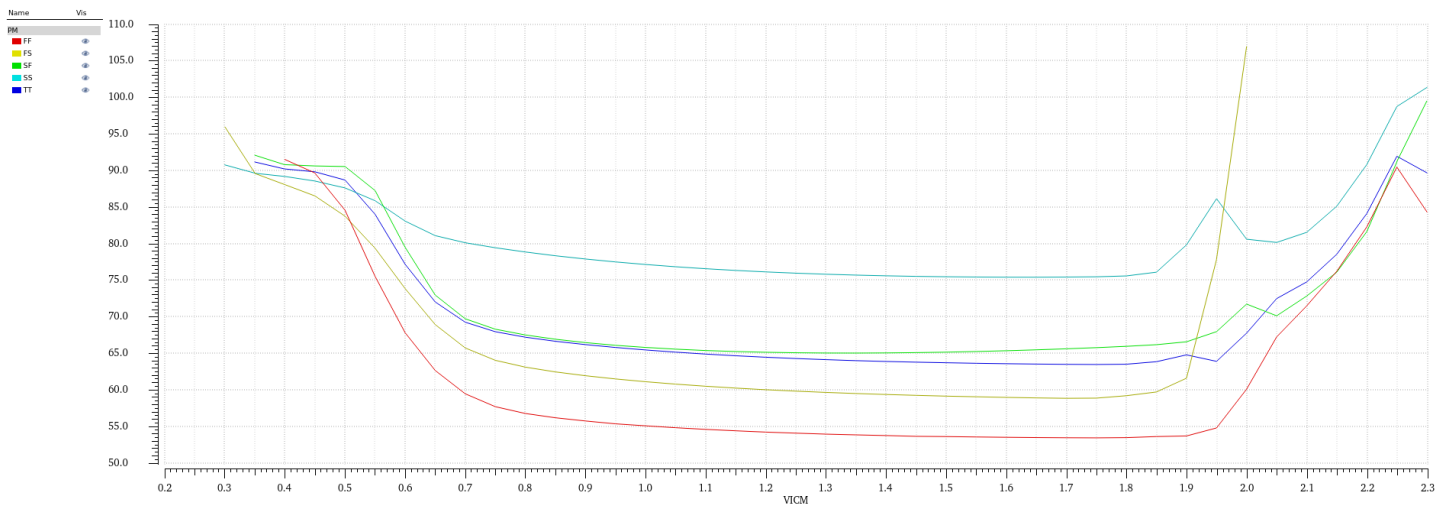
Loop gain: $\beta = 1 \Rightarrow LG = A_{ol}$



A_{ol} phase

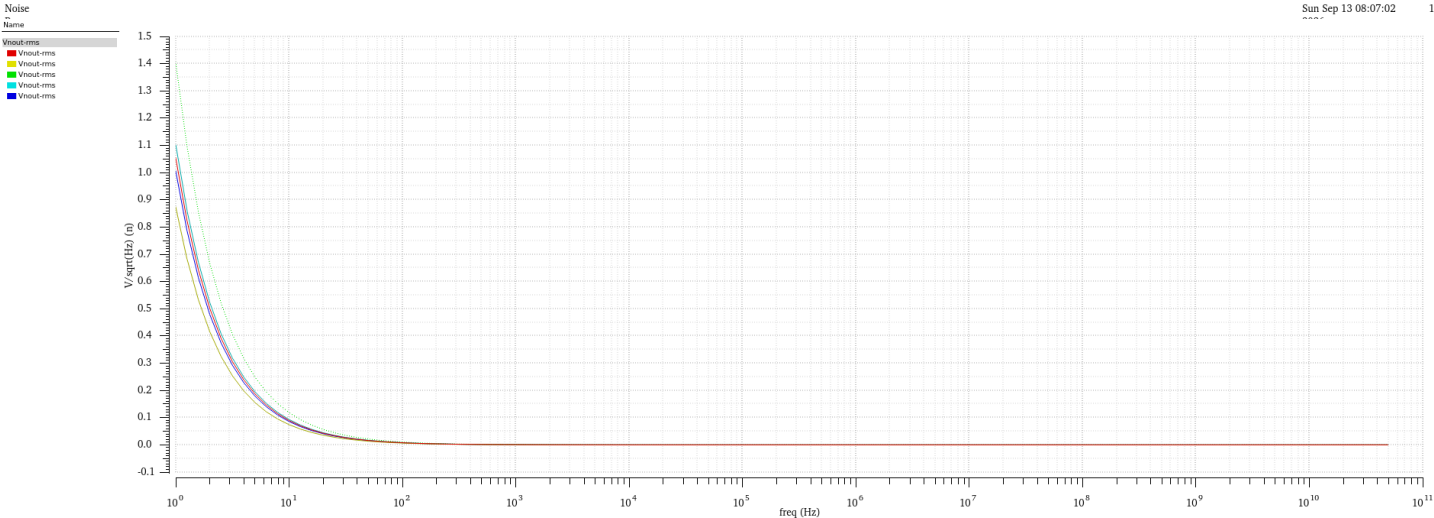


UGF vs V_{ICM}



PM vs V_{ICM}

Noise analysis



$V_{n-out-rms}$

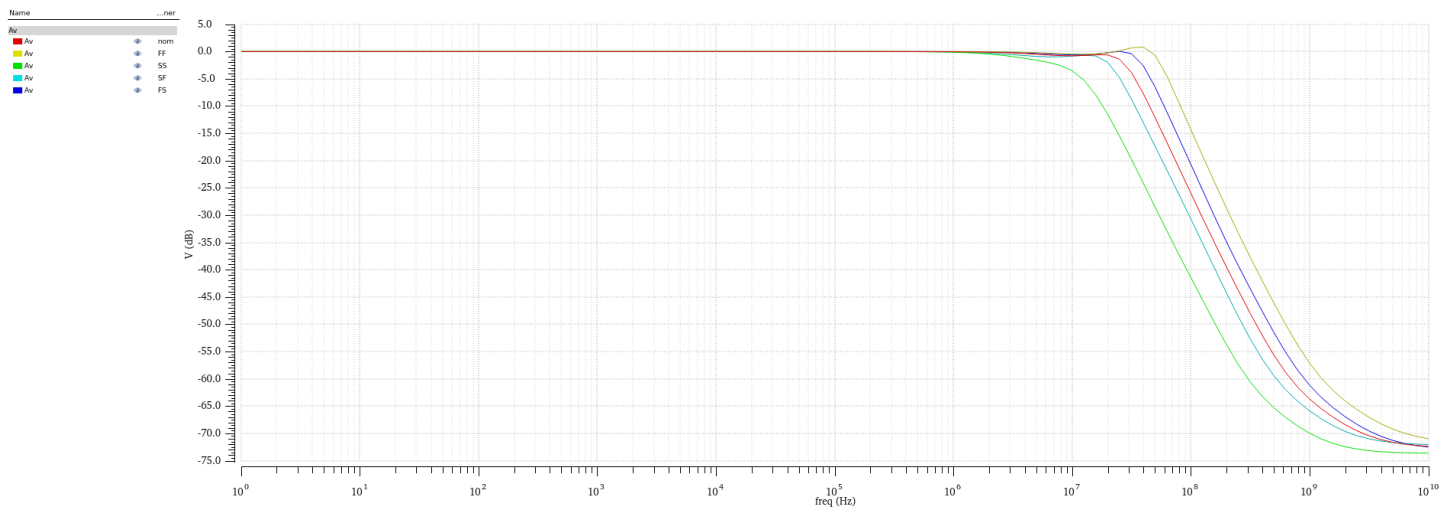
Window Expressions Info Help

cadence

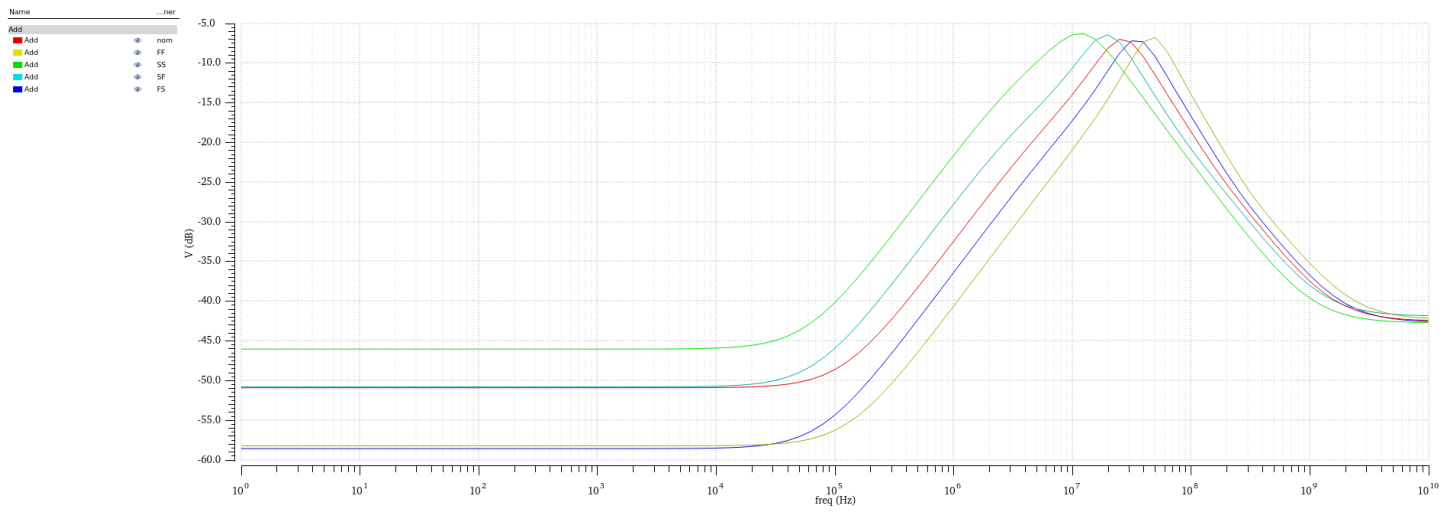
Device	Param	Noise Contribution	% Of Total
/I0/M35	id	3.05102e-08	35.08
/I0/M34	id	2.62315e-08	30.16
/I0/M34	fn	6.64965e-09	7.64
/I0/M19	id	5.57745e-09	6.41
/I0/M14	id	4.90693e-09	5.64
/I0/M35	fn	2.75612e-09	3.17
/I0/M7	id	1.33911e-09	1.54
/I0/M13	id	1.14801e-09	1.32
/I0/M14	fn	9.46255e-10	1.09
/I0/M3	id	9.36273e-10	1.08
/I0/M2	id	8.37513e-10	0.96
/I0/M0	id	7.09892e-10	0.82
/I0/M1	id	6.59272e-10	0.76
/I0/M33	id	5.30236e-10	0.61
/I0/M6	id	4.87597e-10	0.56
/I0/M12	id	4.23942e-10	0.49
/I0/M19	fn	3.5354e-10	0.41
/I0/M13	fn	2.24829e-10	0.26
/I0/M4	id	1.95709e-10	0.22
/I0/M27	id	1.35253e-10	0.16

Integrated Noise Summary (in V^2) Sorted By Noise Contributors
Total Summarized Noise = 8.69831e-08
Total Input Referred Noise = 3.53649e-05
The above noise summary info is for noise data

Noise summary



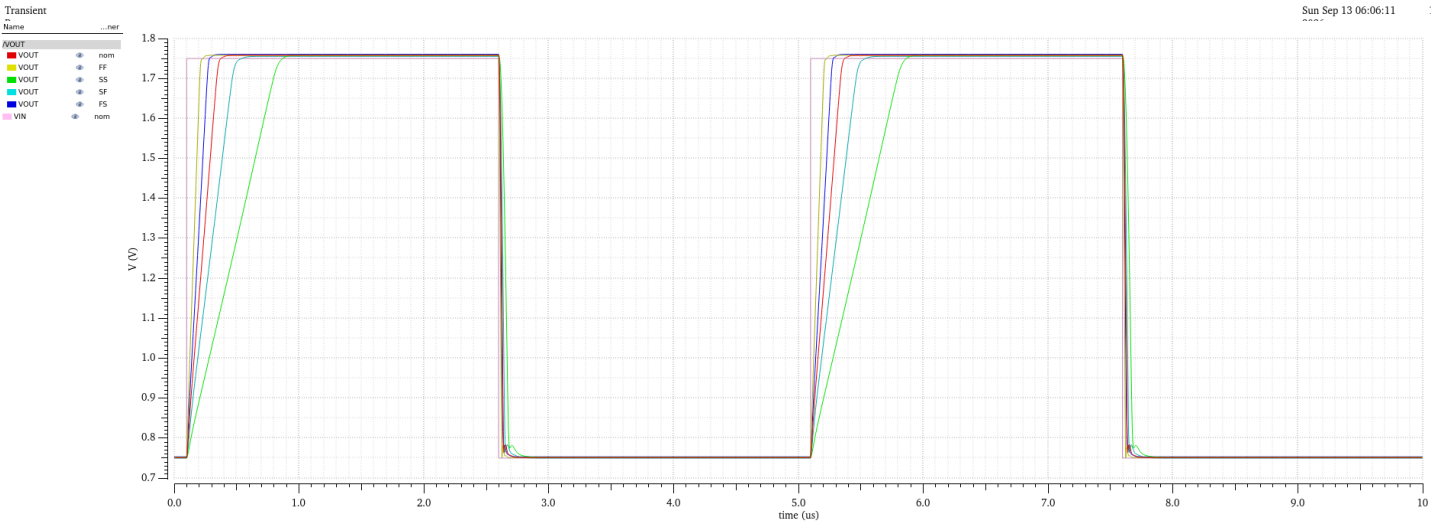
A_{cl} dB



A_{dd} dB

Transient analysis

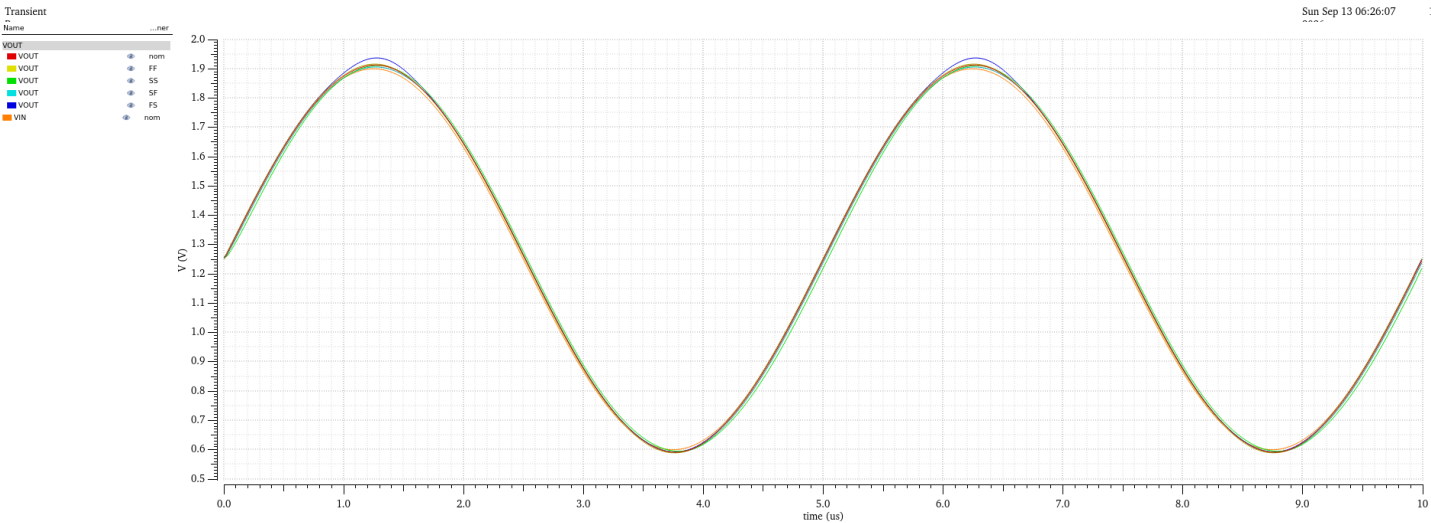
Applying 200KHz – 1V pk-to-pk pulse signal



Vout response for the corners

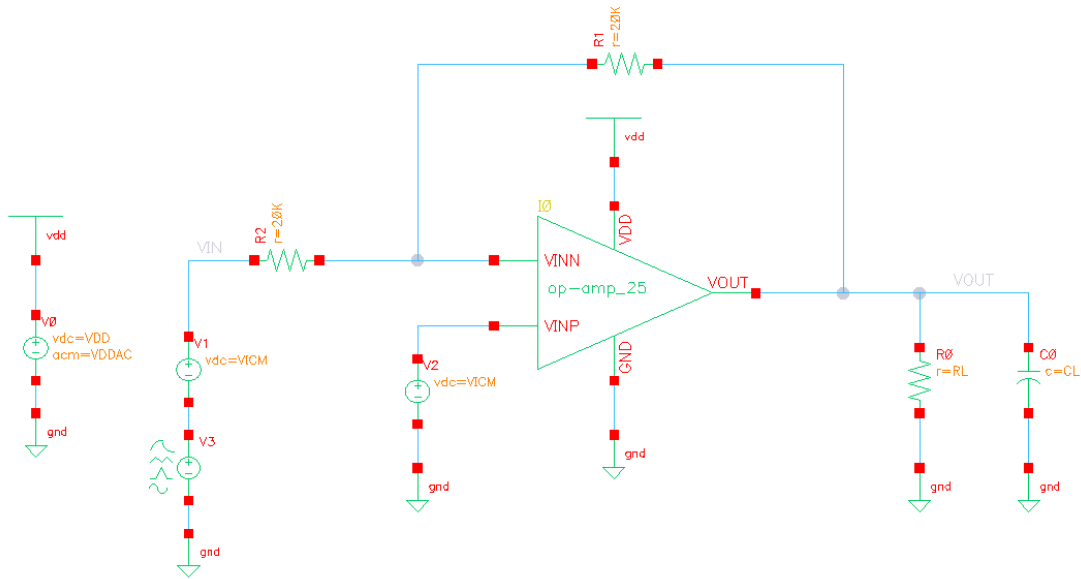
	TT	FF	SS	FS	SF
SR +	3.86 MA/F	1.353 MA/F	1.353 MA/F	5.668 MA/F	2.526 MA/F
SR –	14.38 MA/F	43.68 MA/F	7.013 MA/F	16.33 MA/F	10.56 MA/F

Applying 200KHz – 1.3V pk-to-pk sin signal



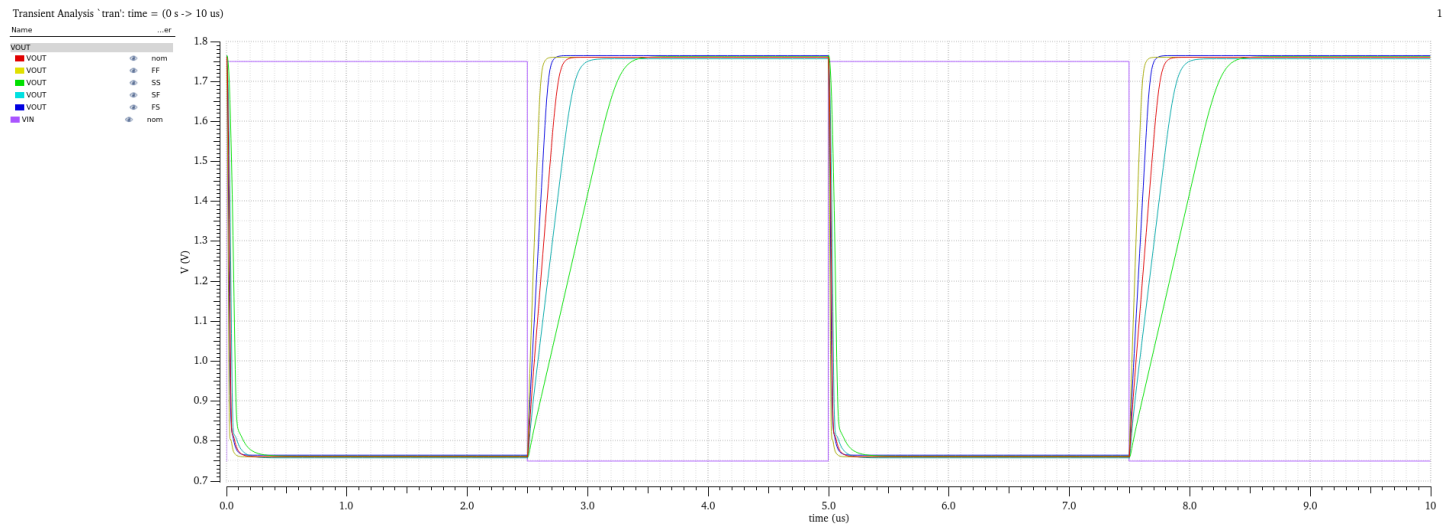
The output perfectly follows the input across the corners

Inverting amplifier topology



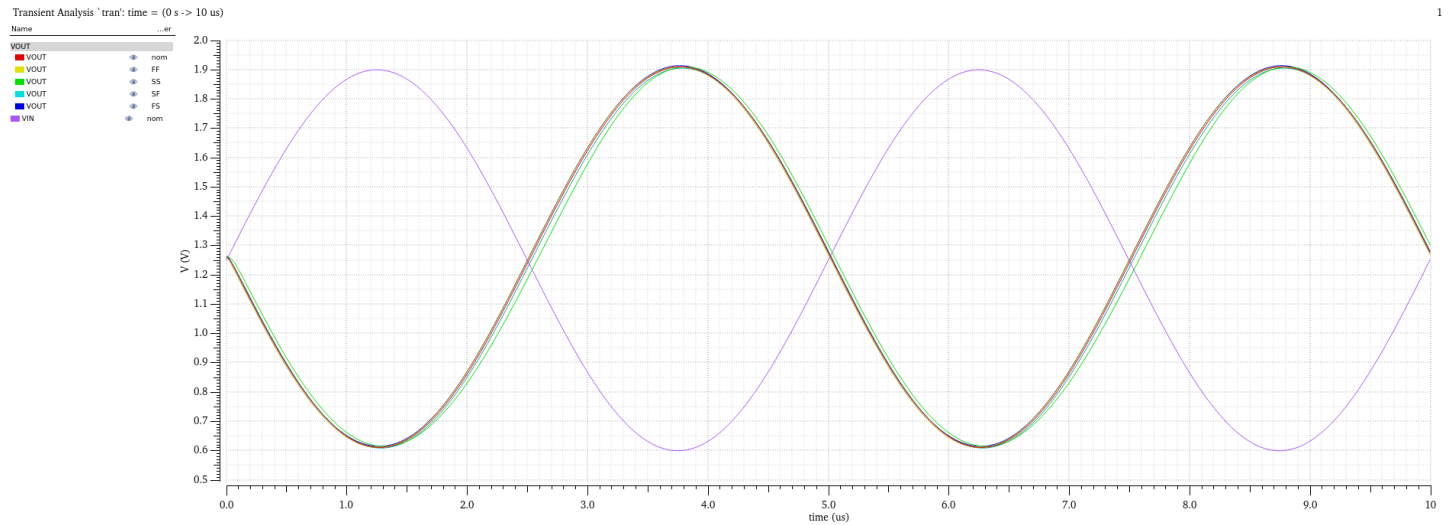
$$V_{out} \approx -\frac{R_1}{R_2} \times V_{in} = -V_{in} \Rightarrow 180^\circ \text{ phase shift with the same amplitude}$$

Applying 200KHz – 1V pk-to-pk pulse signal



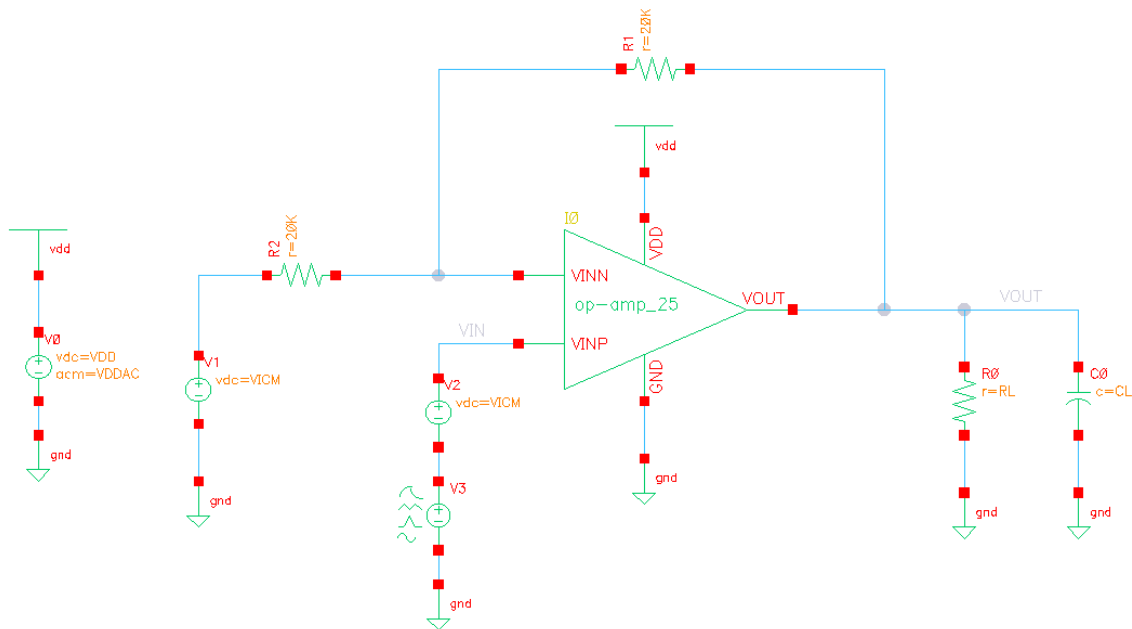
The output response across the corners

Applying 200KHz – 1.3V pk-to-pk sin signal



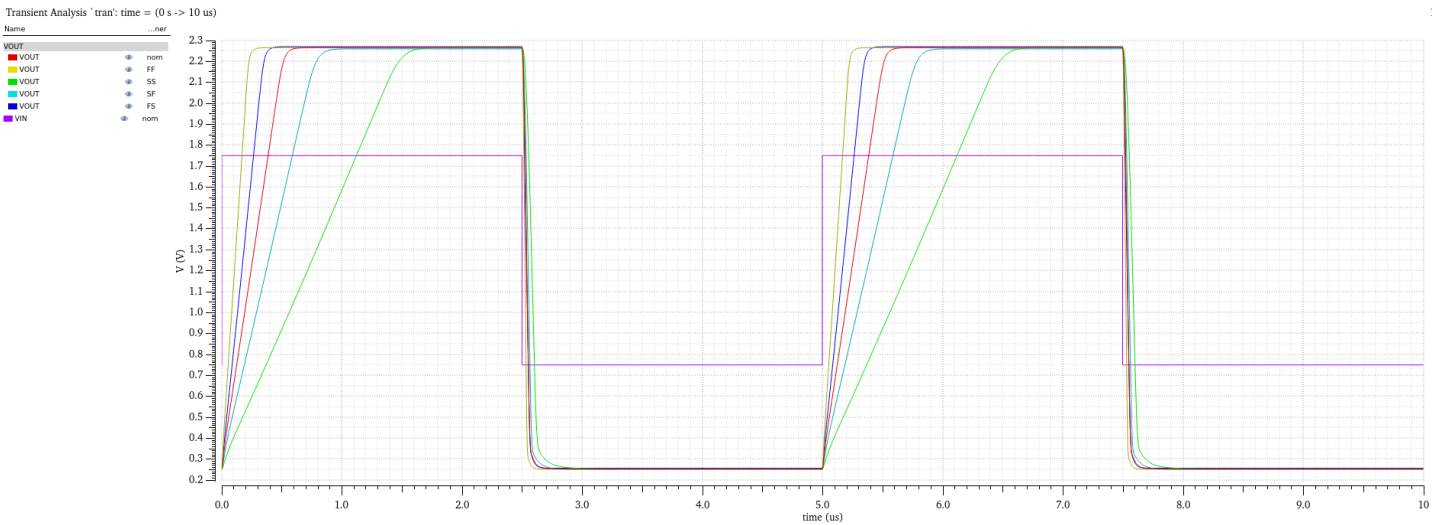
The output is a 180° shifted version of the input across the corners

Non-inverting amplifier topology



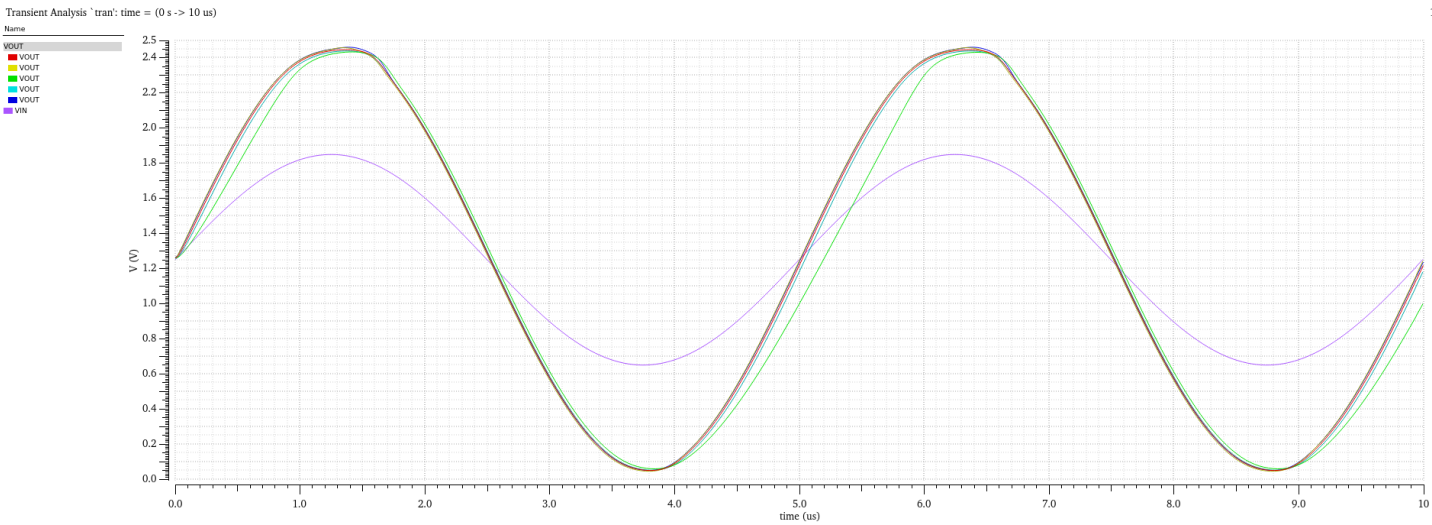
$$V_{out} \approx \left(1 + \frac{R_1}{R_2}\right) \times V_{in} = 2V_{in} \Rightarrow \text{the output in phase with the input with doubled amplitude.}$$

Applying 200KHz – 1V pk-to-pk pulse signal



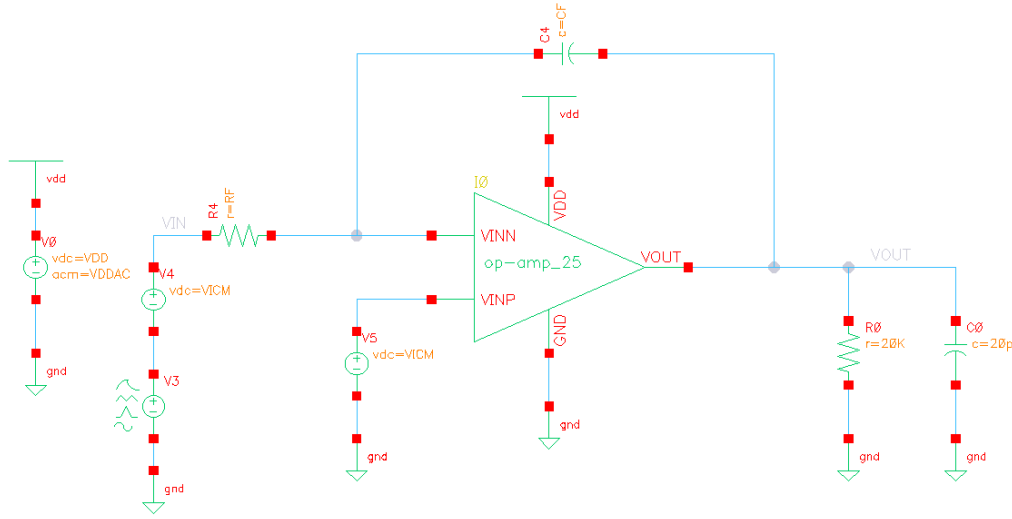
The output response across the corners

Applying 200KHz – 1.2V pk-to-pk sin signal



The output is doubled version of the input and in phase with it

Integrator amplifier topology



$$V_{out} = -\frac{1}{R_F C_F} \int V_{in} dt$$

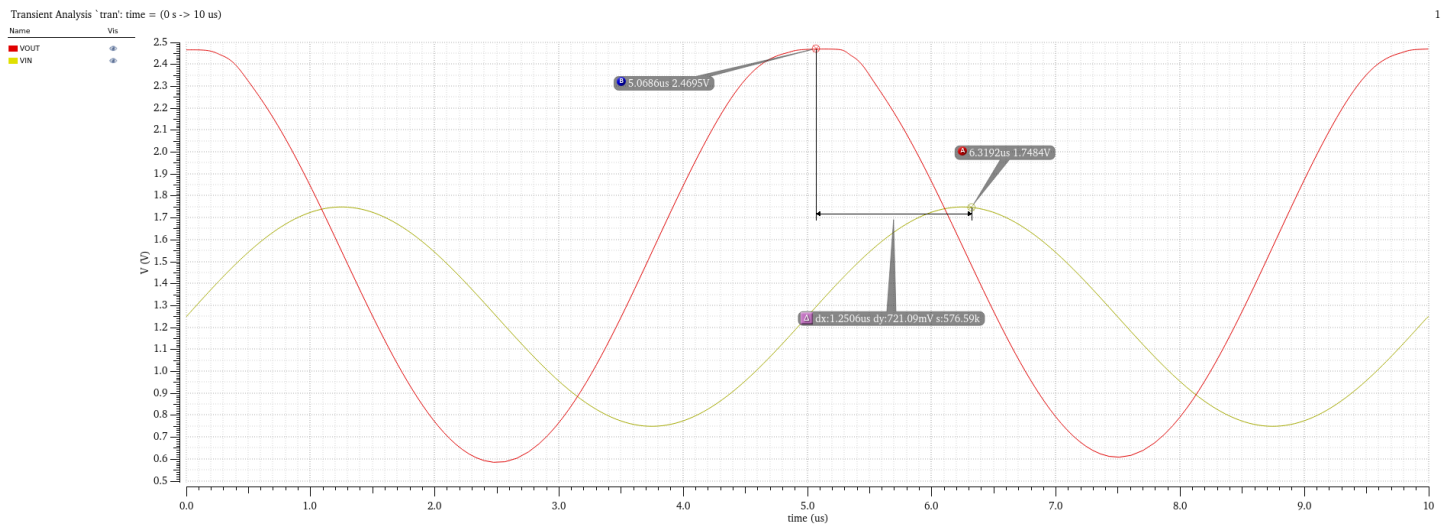
For sin input, $V_{out} = c_1 \cos(2\pi f \times t) + c_2 \Rightarrow$ the output is 90° shifted signal from the input with c_1 gain and c_2 DC shift.

Applying $200\text{KHz} - 1\text{V}$ pk-to-pk sin signal, $R_F = 20\text{K}\Omega$, $C_F = 20\text{pF}$

$$V_{out} = -\frac{1}{20\text{K}\Omega \cdot 20\text{pF}} \int \sin(2\pi \cdot 200\text{KHz} \times t) dt = \frac{1}{20\text{K}\Omega \cdot 20\text{pF} \times 2\pi \cdot 200\text{KHz}} \cos(2\pi \cdot 200\text{KHz} \times t) + c$$

$$\approx 2 \cos(2\pi \cdot 200\text{KHz} \times t) + c$$

the output is 90° shifted signal from the input with 2 gain and c DC shift.



The time shift between the output and the input signals is $\Delta t = 1.25\mu\text{s} \Rightarrow$ the phase shift ϕ can be getting by $\frac{\Delta t}{T} \times 2\pi$

$$\phi = \frac{1.25\mu\text{s}}{5\mu\text{s}} \times 2\pi = \frac{\pi}{2} \equiv 90^\circ$$

Conclusion

Working through this project, from hand calculations to final corner simulations, gave a practical view of how the choice of cascode biasing affects the overall trade-off between compactness and robustness.

Comparing the two implemented schemes showed that sharing a single cascode mirror keeps the design small and power-efficient, but its bias points are more sensitive to process variation. Splitting the biasing into independent branches per node costs extra area and current, but pays off in steadier performance when the process shifts across corners.

Across both approaches, the closed-loop tests (inverting, non-inverting, and integrator) behaved exactly as predicted by theory, which builds confidence that the core topology and sizing methodology were sound. That said, a few corner cases landed just short of the target specs, which points to some headroom left for further tuning of the compensation network and device sizing before the design would be considered fully corner-robust.